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When Losses Turn into Loans: The Cost of Weak Banks†

By Laura Blattner, Luisa Farinha, and Francisca Rebelo*

We provide evidence that banks distort the composition of credit supply in order to comply with ratio-based capital requirements in times of economic distress. An unexpected intervention by the European Banking Authority provides a natural experiment to study how banks respond to falling below minimum required capital ratios during an economic downturn. We show that affected banks respond by cutting lending but also by reallocating credit to distressed firms with underreported loan losses. We develop a method to detect underreported losses using loan-level data. The credit reallocation leads to a reallocation of inputs across firms. We calculate that the resulting increase in input misallocation accounts for about 22 percent of the decline in productivity in Portugal in 2012. (*JEL* E23, E32, G21, G28, G32, G38)

The cornerstone of modern financial regulation consists of rules governing bank indebtedness. Restrictions on bank indebtedness most commonly take the form of a minimum ratio of equity to risk-weighted assets (RWA). When a bank incurs losses, for example in the wake of a financial crisis, the bank's equity cushion absorbs the losses, which may cause the bank to drop below its required regulatory minimum. Understanding how banks adjust to comply with minimum ratio requirements after incurring significant losses is crucial to evaluating how well ratio-based capital requirements perform as a policy tool.

Existing research has shown that distressed banks frequently cut back the supply of credit instead of rebuilding their equity cushion. However, little attention has been paid to how banks adjust the composition of credit supply in order to comply with minimum capital ratios in distressed conditions. At the same time, a growing body of evidence has highlighted the link between input misallocation and slow productivity

* Blattner: Stanford University, Graduate School of Business (email: lblattner@stanford.edu); Farinha: Banco de Portugal, Research Department (email: lfarinha@bportugal.pt); Rebelo: Boston College, Caroll School of Management (email: rebelof@bc.edu). Emi Nakamura was the coeditor for this article. We thank four anonymous referees for many helpful suggestions, as well as Anat Admati, Juliane Begenau, Gabriel Chodorow-Reich, John Coglianese, Andrew Garin, Gita Gopinath, Jeff Frieden, Sam Hanson, Eben Lazarus, Matteo Maggiori, Pascal Noel, Martin Rotemberg, Jesse Schreger, Jann Spiess, Jeremy Stein, and Adi Sunderam for helpful comments. The authors also thank Nuno Alves, Antonio Antunes, Diana Bonfim, Isabel Correia, Gil Nogueira, Maximiano Pinheiro, and members of the research team at the Bank of Portugal. The contribution by Laura Blattner to this paper has been prepared under the Lamfalussy Fellowship Program sponsored by the European Central Bank. Any views expressed are only those of the authors and do not necessarily represent the views of the ECB, the Banco de Portugal or the Eurosystem.

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growth. However, this literature has not linked input misallocation to credit misallocation induced by distorted incentives of distressed banks.

In this paper, we show that adjustments to ratio-based capital requirements imposed in the aftermath of the European sovereign debt crisis heightened input misallocation by reinforcing distorted lending incentives of distressed banks. To establish this causal chain, we exploit an intervention by the European Banking Authority (EBA) in 2011, which caused a subset of banks to be below their minimum regulatory capital ratios. We show that affected banks responded by further distorting their reporting and lending choices at the micro level. We then show how these distorted incentives at the micro level drive the misallocation of inputs across firms and aggregate up to large negative effects on productivity at the macro level.

We establish the first link in the causal chain by exploiting quasi-experimental variation in banks' minimum regulatory capital ratios. The EBA in 2011 unexpectedly announced that a subset of European banks had to meet certain capital ratios by mid-2012, which substantially affected a subset of Portuguese banks. Our exposure definition exploits both eligibility, which was based on bank size, and the severity of the regulatory capital shortfall, which was determined by prior sovereign bond holdings.[1]

We complement the quasi-experimental variation in banks' regulatory capital ratios with a method to detect when banks delay the reporting of loan losses on nonperforming loans. In the period we study, banks are required to deduct a loss when a borrower falls behind on loan repayments. In Portugal, there are regulatory rules that tie the size of this mandatory deduction to the length of time a borrower has been behind on repayment.[2] These rules imply that when reporting on an individual basis, banks can reduce loan losses required under these regulatory rules by underreporting the time a firm has been behind on repayment.[3] We develop an algorithm to measure loss underreporting in monthly bank reports on the same firm. Because the regulatory deduction schedule features several discrete jumps, the incentive to underreport is largest just below a jump. We show that these incentives lead to bunching in the data, providing compelling evidence that banks strategically report to minimize losses.

Our main result, which establishes the first link in the causal chain, is that exposed banks respond to higher ratio-based capital requirements not only by cutting back on lending but also by reallocating credit to a subgroup of distressed firms whose loan losses banks had been underreporting prior to the EBA announcement. In contrast, exposed banks do not increase credit to distressed firms that are not underreported. These results are estimated in a difference-in-difference design, in which we compare

[1] Defining exposure only based on eligibility would imply that we compare big and small banks. In addition, this approach would reduce statistical power since not all eligible banks were affected by the EBA exercise. We confirm that both groups of banks, based on our exposure definition, are balanced on observables (though some moderate size imbalance remains) and that sovereign bond holdings do not follow differential trends prior to the EBA announcement.

[2] As in all European countries, the reporting of loan losses has to follow international accounting standards (IAS39). In addition, nonconsolidated/individual statements, which are relevant for tax purposes, follow the rules for impairment losses set by a Portuguese rule, Notice 3/95. This rule was repealed in 2015. Notice 3/95 ties the size of the impairment loss to the number of months a loan has been overdue as well as the type of collateral.

[3] Underreporting in this context is only defined relative to the specific Notice 3/95. Since the financial supervision authorities had the right to exercise discretion in how to apply Notice 3/95, our findings do *not* imply that banks misreported losses.

changes in credit from exposed and nonexposed banks to the same firm. We show that this credit reallocation is unlikely to be driven by increased credit demand from underreported firms. Exposed banks change their credit allocation only in the period between the EBA announcement and the EBA deadline. Firm-level shocks driving up credit demand would hence have to match the exact timing of the regulatory intervention to be able to account for our results. Moreover, given that we compare changes in lending to the same firm, firm-level shocks would have to drive up credit demand at exposed but not at nonexposed banks. To lend further credibility to our results, we show that underreported firms borrowing from exposed and nonexposed banks do not have diverging pre-trends in credit or liquidity, that observable measures of firm quality are not correlated with the borrowing share from exposed banks, and that our results are robust to controlling for relationship characteristics such as whether the bank is the main lender.

A natural explanation for the observed changes in credit composition is that the shortfall to the ratio-based requirements heightens distorted lending incentives for exposed banks in these distressed conditions. These incentives are likely driven by banks delaying the recognition of losses, which would adversely affect their regulatory capital ratios. We show that banks had been underreporting loan losses with the onset of the European sovereign debt crisis in 2010. While this underreporting allowed banks to boost reported capital ratios, it also locked banks into a vicious cycle with financially distressed firms whose losses have not yet been fully accounted for on banks' financial statements. Cutting lending to an underreported firm runs the risk of pushing that firm into bankruptcy, which would force the bank to recognize previously underreported losses. The ratio-based capital requirements imposed by the EBA give exposed banks an additional reason to avoid capital-reducing losses and to roll over loans to underreported firms. Consistent with this incentive to delay losses, we find that exposed banks sharply increase the amount of loss underreporting for the duration of the EBA intervention. We also test whether our results could be explained by risk-shifting incentives at exposed banks. However, we find no evidence that exposed banks increase risky lending to new or existing clients during the EBA intervention.

We establish the second link in the causal chain by showing how the changes in credit composition affect firms' input use. We first run a firm-level version of our firm-bank specification to confirm that firms do not undo the firm-bank level credit shocks by substituting among different lenders. In the next step, we estimate the effect of the credit shock on input use by instrumenting for the firm-level credit shock with the firm-level pre-intervention borrowing share from exposed banks. The credit shock, which is positive for underreported firms and negative for all other firms, has a large and significant effect on input use. Underreported firms see a relative increase in capital and labor by 9 percent and 6 percent respectively, while all other firms cut back on capital and labor by 8 percent and 6 percent respectively.

In the final step of the causal chain, we show that the credit reallocation to firms with underreported losses negatively affects aggregate productivity growth. Based on the popular approach of measuring wedges in the first-order condition for inputs (Restuccia and Rogerson 2008; Hsieh and Klenow 2009), we show that the marginal output gain from allocating more inputs to underreported firms is much smaller relative to nondistressed firms in the economy. We use a decomposition of aggregate productivity growth based on Osotimehin (2019) to aggregate our firm-level estimates. A simulation exercise based on this decomposition suggests that the credit reallocation to underreported firms accounts for about 22 percent of the productivity decline in 2012. The simulation exercise shows that the magnitude of the effect depends on the extent to which the reallocation of credit causes the dispersion of wedges across firms to increase. Dispersion is driven both by preventing inputs held by underreported firms from being reallocated to firms where these inputs would have earned higher returns and by shrinking the available credit supply for firms with potentially high input returns, forcing them to shed resources.

Our work is related to a body of literature documenting how ratio-based capital requirements can lead distressed banks to cut back the supply of credit rather than raising new equity (Peek and Rosengren 1995; Anjan 1996; Behn, Haselmann, and Wachtel 2016; Koijen and Yogo 2015; and Gropp et al. 2018). While we confirm the finding on credit supply, our primary contribution lies in documenting the effects on credit composition arising from distorted lending incentives and the resulting effects on misallocation and productivity.

While this paper is primarily about the effects of regulatory capital deficiencies, our results can shed light on the distortions caused by a distressed banking system more generally. Our results are consistent with theoretical predictions for the distorted incentives of distressed banks (Admati et al. 2018) as well as empirical evidence of "zombie" lending of distressed banks in Japan (Peek and Rosengren 2005; see Sekine, Kobayashi, and Saita 2003 for a survey). The "zombie" lending phenomenon has received renewed interest following Europe's experience since 2008. This literature has provided reduced-form evidence for evergreening in Europe but not established causality (Schivardi, Sette, and Tabellini 2022; Albertazzi and Marchetti 2010; and Acharya et al. 2019). Beyond introducing quasi-experimental variation in bank capital adequacy to establish causality, we more precisely estimate the extent of "zombie" lending by relying on our underreporting measure instead of measures of poor firm performance. Our work ties the "zombie" lending literature with research on the real effects of this phenomenon. So far, there has been no conclusive evidence on how costly distorted lending is for the economy. Existing research provides evidence that the continued existence of "zombie" firms can have negative spillovers on healthy firms in the same industry (Caballero, Hoshi, and Kashyap 2008; Adalet McGowan, Andrews, and Millot 2018; and Acharya et al. 2019).[4] We take a much more direct approach and show how credit distortions drive the misallocation of resources, which in turn lowers aggregate productivity.

Finally, our work links existing research on bank shocks and credit supply disruptions with a growing body of literature that argues that the misallocation of inputs is a key cause of low productivity (Restuccia and Rogerson 2008; Hsieh and Klenow 2009). While there is a growing body of theoretical models that show how firm-level financial frictions can drive input misallocation (Gopinath et al. 2017; Moll 2014; and Midrigan and Xu 2014), this causal mechanism has not been documented in a quasi-experimental setting. This paper fills this gap by combining quasi-experimental variation in the banking sector that drives credit misallocation

[4] Schivardi, Sette, and Tabellini (2022) however find no such effects in Italy.

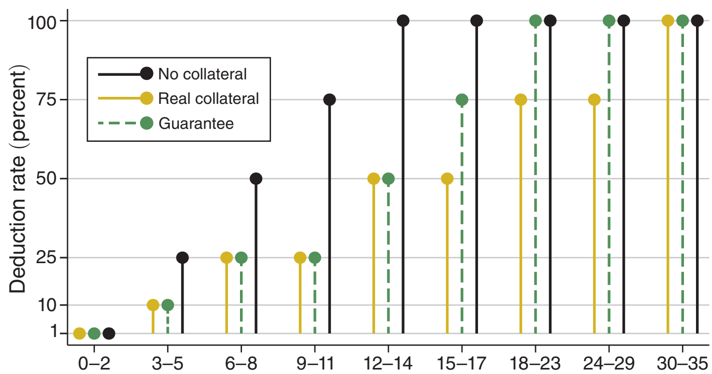

Figure 1. Regulatory Deduction Schedule for Loan Losses

Notes: This graph shows the regulatory rules according to Notice 3/95 that govern mandatory minimum deductions for loan losses based on the number of months a loan has been overdue and the type of collateral. The *x*-axis shows the number of months a loan has been overdue.

with matched firm-level data on real outcomes to establish a causal chain between credit (mis)allocation and input (mis)allocation.

The remainder of the paper is organized as follows: Section I describes our method for measuring loss underreporting. Section II describes the natural experiment, the data and our results. Section III quantifies the effects on aggregate productivity. Section IV concludes.

I. Loss Underreporting: Measuring Distorted Lending Incentives

This section provides background on the regulatory environment that governs the reporting of loan losses in Portugal and describes our methodology for measuring the underreporting of loan losses. We demonstrate that our method produces reliable results by showing that underreporting responds to incentives present in the regulatory rules.

A. *Loan Loss Reporting in Portugal*

We exploit the rules that govern loan impairment losses in Portugal to measure the underreporting of loan losses.[5] In the period we study, loan impairment losses are subject to international accounting standards (IAS39).[6] In addition, nonconsolidated/individual statements, which are relevant for tax purposes, followed the rules for impairment losses set by a Portuguese rule, Notice 3/95. Notice 3/95, which we exploit to identify underreporting, tied the size of the impairment

[5] An impairment loss is an expense that a bank deducts from its income to reflect that a loan may not get repaid in full. Impairment losses reduce banks' regulatory capital by reducing retained earnings. On the balance sheet, impairment losses mark down the value of an asset.

[6] IAS39 requires banks to deduct the difference between the book value of the loan and discounted expected cash flows.

loss to the number of months a loan has been overdue as well as the type of collateral (see Figure 1).[7]

We exploit the detailed reporting of overdue loans by banks to measure loss underreporting. Banks are required to report the length a loan has been overdue, as well as the type of collateral, to the Central Credit Register (*Central de Responsabilidades de Credito*) at a monthly frequency.[8] Banks report the time overdue in discrete intervals, or buckets, which correspond to the regulatory buckets in Notice 3/95 shown in Figure 1.

We focus on firm-finance loans granted to nonfinancial firms. Firm-finance loans tend to have longer maturities than some other credit products, such as credit cards, and therefore are better suited for detecting overdue credit underreporting which requires us to track a lending relationship over time. Firm-finance loans constitute the main loan product for firms and capture about 36 percent of the banks' corporate loan portfolio. As the vast majority of firms have at least one firm-finance loan with each of their lenders, we capture almost the entire population of bank-dependent firms in Portugal. Table A1 in the online Appendix presents descriptive statistics on the loans that we use to measure the underreporting of loan losses. Seventy-three percent of loans are collateralized and 67 percent have an origination maturity above a year.

B. *A Method to Detect Underreporting of Loan Losses*

Our aim is to measure to what extent banks underreport loan losses by managing the reported time a loan has been overdue. Unfortunately, we cannot simply compare reported time overdue to the actual time overdue in the data since banks do not provide identifiers to track loans over time. Instead, we work with a monthly firm-bank panel, with the overdue loan balance reported separately for each overdue category ("bucket").

Algorithm.—We develop an algorithm to measure the extent of underreporting in each reporting bucket for all firm-bank pairs at a monthly frequency. We denote the observed loan balance reported in overdue bucket k in month t by $B_{ib}(t;k)$ where i denotes the firm and b the bank. We drop the firm-bank subscripts in the discussion that follows. There are 14 reporting buckets which correspond to the overdue buckets in the regulatory schedule: $k \in \{\{0\},\{1\},\{2\},\{3,4,5\},\ldots,\{30,\ldots,35\}\}$.

The goal of the algorithm is to measure excess mass, a term we borrow from the bunching literature.[9] We define excess mass in an overdue bucket k in month t,

[7] While the Portuguese financial supervisor ensures that banks comply with international accounting standards at a consolidated basis, Notice 3/95 sets an easily observable benchmark for loan losses at an individual level, and banks avoid large discrepancies between IAS39 and Notice 3/95.

[8] Banks start reporting this variable in 2009.

[9] Our setup differs from the standard bunching setting where the researcher observes a continuous variable, such as house prices or test scores. In those settings, bunching can be measured based on excess mass in the observed cross-sectional distribution at points of particular importance, such as test score cutoffs (see Diamond and Persson 2016; Dee et al. 2019; or Best and Kleven 2017). In our setup, we instead calculate excess mass from repeated observations of the same firm-bank unit and detect discrepancies in observed reporting for the same firm-bank pair over time. In contrast to the standard setting, we also have to address the challenge that reported time is not continuous but discretized.

TABLE 1—EXAMPLES OF LOAN LOSS UNDERREPORTING

EUR million	< 30 days	Overdue 1 month	2 months	Repayment	Excess mass
Panel A. Examples 1 to 3					
2012 million 1	5			0	0
2012 million 2		5		0	0
2012 million 3		5		0	5
2012 million 4		3		2	3
Panel B. Example 4 and 5					
2012 million 1		5		0	0
2012 million 2	5	5		0	5
2012 million 3		7		3	3.5
2012 million 4		9		3	4
2012 million 5			9	0	0

Notes: This table shows stylized examples of the loan data collapsed to the monthly firm-bank level to illustrate the algorithm we use to measure underreported losses. We show lending volumes of a hypothetical firm-bank pair. The first three columns show the first three reporting buckets corresponding to how long a loan has been overdue. The fourth column shows repayment amount and the fifth column shows the excess mass, the amount of overdue loans that is underreported. Panel A illustrates examples 1–3 explained in Section I, and panel B illustrates the remaining two examples.

$E(t;k)$, as the lending balance that is reported in a bucket k that exceeds the lending balance we would have expected to observe in bucket k based on the amount observed at $t-1$. For the first three reporting categories, which consist of a single month, excess mass is defined as $E(t;k) = B(t;k) - B(t-1;k-1)$. Intuitively, the loan balance we observe in bucket k at t must be the loan balance that has moved up from the preceding bucket in the previous period. We define excess mass as the deviation from this identity. For reporting buckets that consist of several months, we have to adjust this simple formula and introduce an auxiliary step, which is described in Appendix A.

We illustrate the algorithm through a series of examples displayed in Table 1, which provides a stylized example of the firm-bank credit panel. The full description can be found in Appendix A.

EXAMPLE 1: *Rows 1 and 2 in panel A of Table 1 show an example where the bank reports correctly. The loan balance of €5 million is reported as overdue less than 30 days in January and then reported as 1 month overdue in February. In this case, we record no excess mass since* $E(2012\text{m}2;1) = B(2012\text{m}2;1) - B(2012\text{m}1;0) = 5 - 5 = 0$.

EXAMPLE 2: *In row 3, we illustrate the first of three mechanisms that banks use to achieve underreporting. The bank does not update the reported time overdue on the €5 million balance. Our algorithm now records excess mass:* $E(2012\text{m}3;1) = B(2012\text{m}3;1) - B(2012\text{m}2;0) = 5 - 0 = 5$.

EXAMPLE 3: *In row 4, we illustrate the effect of a repayment. The firm repays €2 million of the €5 million overdue balance and the bank continues to report this balance to be only 1 month overdue. Outflows of these kind pose no problem and the algorithm will correctly record excess mass equal to €3 million:* $E(2012\text{m}4;1) =$

$B(2012\text{m}4;1) - [B(2012\text{m}3;0)] = 3 - 0 = 3$. *A potential concern is that there may be cases in which, each month, the firm repays an overdue installment but a new one falls overdue. However, in such cases we should observe a reduction in the performing credit balance over time as the firm slowly pays down the loan. Yet for 70 percent of observations that feature underreporting, the performing credit balance remains unchanged. Moreover, we present several validity checks that suggest that underreporting responds to the incentives inherent in the regulatory rules which is inconsistent with underreporting being driven by regular accounting practices.*

In general, flows (such as repayments, a restructuring, write-off, or new loan installments falling overdue) require us to adjust the basic formula above to the following expression:

$$(1) \qquad E(t;k) = [B(t;k) - IN(t;k)] - [B(t-1;k-1) - OUT(t;k-1)],$$

where $IN(t;k)$ denotes inflows (more of the loan balance falling overdue) and $OUT(t;k-1)$ denotes outflows (restructuring, write-offs, or repayments). Inflows will lead us to overestimate excess mass since they add to observed mass in a given bucket and thus need to be subtracted from observed mass.[10] In contrast, outflows from the prior bucket will lead us to underestimate excess mass since we subtract too much past mass and need to be added back.[11]

EXAMPLE 4: *In panel B, we illustrate a case where we need to adjust for outflows. There are two installments of €5 million that fall overdue in two consecutive months. In March, the firm makes a repayment of €3 million. The bank merges the two overdue balances to report a single balance of €7 million as overdue one month. We calculate the outflow the following way. The total change in the overdue balance is −€3 million implying total outflows of €3 million. We then need to decide which of the two balances the firm paid down. Our baseline version attributes the repayment to all buckets with positive balances according to the size of the balance in each bucket. In this case, we would attribute half the repayment to bucket 0 and half the outflow to bucket 1. We test whether the results are sensitive to how we allocate the outflows. Figure A1 in the online Appendix shows that the results do not change much when shifting the repayments to the lowest (highest) buckets. The reason is that repayments are usually small relative to the total balance. The algorithm implies that excess mass in this example is equal to*

$$\begin{aligned} E(2012\text{m}3;1) &= B(2012\text{m}3;1) - [B(2012\text{m}2;0) - OUT(2012\text{m}2;0)] \\ &= 7 - [5 - 0.5 \times 3] = 3.5. \end{aligned}$$

Note that even if we assume that the €3 million were used exclusively to pay down the longest overdue portion of the loan, the bank should have at least reported €2 million as overdue for 2 months (instead of 1 month), leading us to slightly

[10] The exception are inflows into the lowest reporting category.

[11] Note that loan restructuring and write-offs are measured as separate entries in our data.

overestimate underreporting. In the most stringent interpretation of the regulation, the bank should have reported the entire €7 million in the 2 month category leading us to significantly underestimate excess mass.[12] We choose a conservative way of calculating underreporting that is closer to a lenient interpretation of the rules.

EXAMPLE 5: *Finally, in the second to last row of panel B, we present an example where we have both inflows and outflows. We observe a balance of €9 million, corresponding to a net change of €1 million relative to the previous month. If we used the approach from the previous example, we would attribute this as a €1 million repayment in bucket 1 and calculate excess mass equal to €9 million. However, in this example, we can correct for installments of €5 million that fall due every other month. Assuming an inflow of €5 million, we can then deduce that the firm must have made a repayment of €3 million leading us to calculate excess mass equal to*

$$\begin{aligned} E(2012\text{m}4;1) &= \big[B(2012\text{m}4;1) - IN(2012\text{m}4;1)\big] \\ &\quad - \big[B(2012\text{m}3;0) - OUT(2012\text{m}4;0)\big] \\ &= [9-5] - [0-0] = 4. \end{aligned}$$

Based on the discussion above, both implementations would be correct but correspond to more and less stringent interpretation of the regulation. We again choose the less stringent interpretation of the rules to obtain a conservative estimate of underreporting. See Appendix A for details on how we calculate installments in the data.

The last row in panel B illustrates that the algorithm is Markovian and only records inconsistencies relative to $t-1$. That is, once a bank reports something that is consistent with last month's report, we no longer record any excess mass even though the bank may still be behind the actual time a loan is overdue. The reason is that without loan identifiers, we can only record inconsistencies in reporting relative to yesterday but cannot make statements about the "true" length of time an individual loan has been overdue. This property also implies that the algorithm also does not penalize for larger discrepancies between actual time overdue and reported time overdue.

Banks use three mechanisms to adjust the reported time overdue: (i) they do not update the reported time (example 2), (ii) they combine new overdue loan installments with the existing overdue loan balance and report a lower average time overdue (example 4), and (iii) they grant new performing credit in exchange for the repayment of the longest overdue portion of the loan (example 5, assuming the bank supplied €3 million of new lending to enable the firm to repay €3 million). In Figure A2 in the online Appendix, we show that most underreporting is driven by the latter two actions.

[12] Based on conversations with the financial regulator, the strict version was the intended interpretation of the regulation but there appears to be some discretion in how to interpret and enforce the regulation.

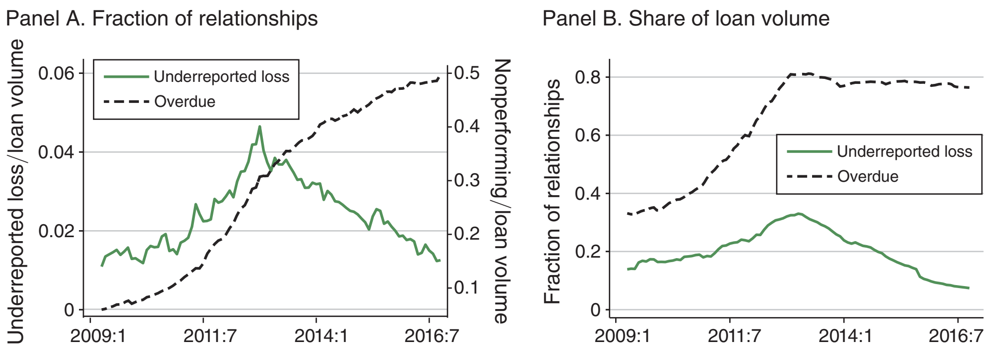

FIGURE 2. PREVALENCE OF LOAN LOSS UNDERREPORTING

Notes: Panel A shows the fraction of firm finance lending relationships that have some overdue loans and the fraction of relationships that are subject to loan loss underreporting as measured by our algorithm. Panel B shows the overdue balance scaled by total loan volume (RHS) and the amount of underreported losses scaled by total loan volume (LHS).

Potential Shortcomings.—We now address two potential shortcomings of using underreporting to identify distorted lending incentives. First, our measure of loss underreporting only applies to firms that already have some overdue loans. It does not capture cases where a bank prevents a firm from falling overdue by granting loans that allow a firm to stay current on loan repayment. However, in the time period we study, a large number of firms have overdue payments in the data (see Figure 2), implying that we capture a large fraction of potential underreporting in the economy. Second, banks can swap out all overdue credit for performing credit. This action will not be captured by the algorithm since there is no more overdue lending reported. However, in our sample period, only about 1 percent of underreported relationships "disappear" in a given quarter.

Validity Checks.—Given that the regulatory deduction schedule features several discrete jumps, we would expect banks to do most of their reporting management in reporting buckets just before a jump ("bunching"). We test whether underreporting in fact occurs in buckets just before a jump. Such responsiveness of bank behavior at the micro level is evidence that our measure is indeed picking up strategic behavior.[13]

We can test this responsiveness by regressing the amount of underreporting in a reporting bucket on the size of the rate increment in the next higher bucket. We run this regression separately for each type of collateral since the regulatory rules differ by collateral type. We describe the regression specification in detail in Appendix B.

The regression confirms that, for each type of collateral, the amount of underreporting is statistically significantly higher when there is an increase in the regulatory rate in the next bucket relative to buckets where the regulatory rate stays constant (see Table B1 in Appendix B). Moreover, we find that underreporting is higher if the increment in regulatory deduction rate is higher, suggesting that underreporting

[13] The algorithm does not restrict excess mass to be zero even when there is no increase in the regulatory rate in the next higher reporting bucket.

responds not only the location of the jumps in the regulatory rate but also to the size of the increment.[14]

We also conduct the following placebo test. When we regress underreported losses on the regulatory increments of *another* collateral type, we should not find positive and significant coefficients in categories where only the *other* collateral type features a jump in the deduction rate. Table B1 in Appendix B shows that we find negative coefficients for all three collateral types, suggesting that there is significantly *less* underreporting when only other collateral types feature an increase in the regulatory rate.

II. The Cost of Undercapitalized Banks: A Natural Experiment

This section first describes the regulatory intervention by the EBA. We then briefly describe our data and present our main results.

A. *The 2011 EBA Special Capital Enhancement Exercise*

In October 2011, the EBA[15] announced a special capital enhancement exercise to force banks with large or overvalued sovereign debt exposures to improve their capital ratios by June 2012. The EBA intervention applied to the largest banks in each country based on a cutoff determined by the EBA.[16] The EBA exercise, at least in its full scope, was plausibly unexpected as banks had already undergone a round of EBA stress tests in June 2011.[17] The intervention led to a large capital shortfall for most eligible Portuguese banks since their eurozone sovereign debt holdings were substantial and often valued above market prices in their balance sheets.[18]

We define a bank as exposed to the EBA intervention if it was both subject to the intervention and had a large capital shortfall. The EBA determined the capital shortfall according to the following formula:

$$\frac{\mathit{Core\ Tier\ 1} - \mathit{sovereign\ debt\ buffer}}{\mathit{RWA}} \geq 0.09.$$

The sovereign debt buffer was designed to reflect losses from banks' sovereign debt portfolios. For sovereign debt held in available-for-sale portfolios, the buffer removed the prudential filters that usually insulate bank capital from fluctuations in market prices. For sovereign debt in held-to-maturity portfolios, the buffer reflected

[14] There is one exception where this monotonicity fails: the largest increment for loans with either real collateral or borrower guarantees, which does not feature more underreporting relative to the second-largest increment. This nonmonotonicity arises because loans in the reporting category just below the second-largest jump have to be declared nonperforming, which has additional negative effects beyond increasing the impairment loss. Nonperforming loan ratios are a closely watched indicator of bank health by both the regulator and financial markets giving banks a reason to concentrate their underreporting in lower reporting buckets.

[15] The EBA is an EU agency tasked with harmonizing banking supervision in the European Union.

[16] Banks covered by the EBA exercise had to jointly hold at least 50 percent of the national banking sector as of the end of 2010 (European Banking Authority 2011, https://www.eba.europa.eu/risk-analysis-and-data/eu-capital-exercise).

[17] The *Financial Times* on October 11, 2011 reported that the EBA requirements were "well beyond the current expectations of banks and analysts." See Patrick Jenkins, Ralph Atkins, and Peter Spiegel, "Europe's Banks Face 9% Capital Rule," *Financial Times*, October 11, 2011.

[18] In Portugal four banking groups (containing seven banks) were subject to the capital exercise.

TABLE 2—DESCRIPTIVE STATISTICS: FIRMS AND BANKS

Firms		Banks			
			Not exposed	Exposed	Diff
	(1)		(2)	(3)	(4)
Assets (m)	1.62	Assets (100 bn)	0.42	0.98	0.56
	(6.05)		(0.32)	(0.34)	(0.21)
Employees	13.47	Sovereign bonds	0.04	0.06	0.02
	(114.15)		(0.04)	(0.02)	(0.01)
Total credit (m)	0.52	Loans	0.46	0.49	0.03
	(4.86)		(0.14)	(0.11)	(0.06)
Share NPLs	0.07	NPLs	0.02	0.02	0.00
	(0.22)		(0.02)	(0.01)	(0.00)
Return on assets	0.06	Return on assets	0.00	0.00	0.00
	(0.08)		(0.02)	(0.00)	(0.00)
Sales growth	0.13	Deposits	0.33	0.40	0.06
	(0.48)		(0.17)	(0.13)	(0.07)
Leverage	0.28	Capital ratio	0.10	0.14	0.04
	(0.73)		(0.14)	(0.02)	(0.03)
Current ratio	2.43	Liquid assets	0.01	0.01	0.00
	(4.29)		(0.01)	(0.00)	(0.00)
Cash/assets	0.13	LTRO	0.08	0.08	0.00
	(0.17)		(0.06)	(0.03)	(0.03)
Fixed assets/assets	0.47	Interbank market	0.22	0.13	−0.10
	(0.29)		(0.20)	(0.11)	(0.06)
Observations	144,134		38	7	

Notes: The table shows descriptive statistics for firms and banks in our sample. All variables are measured at the end of 2010. We only include firms in our sample: firms that consistently report to the annual firm census in our sample period in 2008–2011. All bank variables with exception of assets are scaled by total assets. Exposed refers to banks that are exposed to the EBA intervention. Diff refers to the difference in means for exposed and nonexposed banks. Standard deviations are in parentheses. LTRO = long-term refinancing operations. NPLs = nonperforming loans.

the difference between the fair value at September 2011 prices and the values the bank had recorded for these holdings.[19]

Our control group consists of banks that were subject to the EBA intervention but had below median sovereign debt holdings (in the group of large banks) and therefore a small capital shortfall under the EBA intervention. We also include in the control group any commercial bank operating in Portugal not subject to the EBA intervention. We exclude any bank whose foreign parent was subject to the EBA intervention in another European country.

Our identification strategy rests on the assumption that there are no unobserved differences between the two groups of banks that could drive the observed credit allocation during the EBA intervention. Table 2 shows that the groups are balanced on observables prior to the EBA intervention—though some imbalance on size

[19] This variation was more important than the moving the constraint up to 9 percent. In Portugal, all banks were already required to have Core Tier 1 ratios of 8 percent in 2011. Moreover, for exposed banks the average capital shortfall (holding RWA fixed) before including the sovereign buffer was only 14 percent of the total shortfall due to the EBA intervention. This implies that without the additional requirement of the sovereign buffer, exposed banks would have faced only very small capital shortfalls. Therefore, most of our variation is driven by losses induced by the sovereign debt buffer as opposed to a change in the minimum required capital ratio.

remains given the selection criteria of the EBA intervention. In particular, we show that both groups of banks made similar use of the ECB's long-term refinancing operations (LTRO), which were also introduced in late 2011. To further address potential confounding effects from the LTRO operations, we show that our results are robust to controlling for the loans that banks obtained under the LTRO program. Another potential concern is that eurozone sovereign debt holdings may be correlated with unobserved differences across eligible banks. Figure A3 in the online Appendix shows that there are no systematic differences in the eurozone sovereign debt holdings of *eligible* banks both before and during the EBA intervention, suggesting that differences in debt holdings are unlikely to reflect short-run shocks that could also affect credit allocation.[20] Figure A3 in the online Appendix shows that while there was considerable stress in sovereign debt markets during this time, the peak in Portuguese sovereign debt spreads does not match the timing of the EBA intervention. This suggests that events in sovereign debt markets are unlikely to account for our results.

Sources of Distorted Lending Incentives.—The specific design of the EBA intervention potentially heightened two sources of distorted incentives for exposed banks. First, existing research has argued that bank shareholders often resist raising new equity and instead prefer to find other ways to comply with higher capital ratios (Admati et al. 2018). Hence exposed banks have an incentive to boost reported regulatory capital by increasing the intensity of their loan loss underreporting and simultaneously rolling over loans to underreported firms.[21] What makes underreported firms special is that banks had not reported the true extent of losses in the past. Thus if an underreported firm gets cut off from credit and goes bankrupt, the bank would have to report all the losses the bank had previously been underreporting. In contrast, for a correctly reported firm, the bank has set aside reserves to cover the loss and has already taken the hit on its capital. It therefore has less of an incentive to keep that firm afloat purely to avoid having to book further losses.

Second, the promise of a potential government bailout might have led to risk-shifting incentives for exposed banks. Exposed banks could plausibly anticipate that as long as they made a credible attempt to comply with the EBA requirements, the Portuguese government would step in to make up any remaining capital shortfall at the compliance deadline.[22] These expectations were validated when in June 2012, at the EBA compliance deadline, the Portuguese government provided €6 billion of capital in the form of convertible contingent bonds to all exposed banks. The anticipated bailout potentially gave bank shareholders the incentive to gamble for the resurrection of distressed borrowers since the bailout was effectively a government guarantee to cover any loss in June 2012. To the extent that underreporting correlates with the firm risk, risk shifting would lead exposed banks to increase their exposure to underreported firms.

[20] The increase in sovereign debt holdings in both groups at the end of 2011 may be driven by the fact that all large banks purchased sovereign debt to use as collateral to access the LTRO program (Crosignani, Faria-e Castro, and Fonseca 2020).

[21] It is important to note that the EBA requirements applied at an consolidated level while impairment losses under Notice 3/95 applied at an individual level. However, as explained in Section I, banks likely avoided noticeable discrepancies in the loan loss reporting between consolidated and individual statements.

[22] In May 2011, the Portuguese government had received a financial assistance package from the International Monetary Fund and European Financial Stability Facility, which explicitly earmarked €12 billion to recapitalize Portuguese banks.

B. *Data*

We use proprietary administrative data from the Portuguese central bank. We combine quarterly bank balance sheet data with information from the EBA website to determine which banks were eligible for the exercise either directly, or through a foreign parent, and to obtain the capital shortfall due to the EBA intervention. We merge the bank information with the credit register data (*Central de Responsabilidades de Credito*), a loan level database, which covers the universe of lending relationships that exceed €50. We collapse the loan data to the quarterly firm-bank level. We then merge this information with balance sheet and other financial variables for nonfinancial firms. The data comes from the Simplified Corporate Information (*Informacao Empresarial Simplificada*), an annual, mandatory firm census.

We work with three final datasets. First, a quarterly dataset of loan balances at the firm-bank level from 2009 to 2015 spanning 45 banks, 144,050 nonfinancial firms, and 380,286 lending relationships. The dataset covers over 90 percent of loans made in Portugal. Second, we collapse the firm-bank data to a quarterly firm-level dataset covering the same time period and number of firms. Third, we use the annual firm-level information from 2009 to 2015. We drop firms with fewer than two employees or missing information (or negative values) on assets or employees in 2008–2011. The firms in our resulting sample cover 81 percent of sales and 73 percent of assets in Portugal. We winsorize all outcome variables at the 1 percent level separately for each two-digit industry.

C. *Results*

Banks subject to the EBA intervention cut credit for all but the subset of financially distressed firms whose loan losses they had been underreporting prior to the EBA intervention. This credit reallocation is present both at the firm-bank level, controlling for the total change in firm-level credit, and at the firm-level. We show that there is a substantial pass-through of the credit shock into employment and investment spending.

Credit Effects at the Firm-Bank Level.—We run the following difference-in-differences specification at the firm-bank level:

$$
\begin{aligned}
(2)\quad g_{ibt}^{credit} = & \sum_{\tau=-2}^{5} \beta_{\tau}^{treat}\left(period_{\tau} \times exposed_{b}\right) \\
& + \sum_{\tau=-2}^{5} \beta_{\tau}^{period}\tau\left(period_{\tau} \times underreported_{ib}\right) \\
& + \sum_{\tau=-2}^{5} \beta_{\tau}^{treatgroup}\left(period_{\tau} \times underreported_{ib} \times exposed_{b}\right) + \theta_{it} + \varphi_{b} \\
& + \beta_{1}^{base}\left(underreported_{ib} \times exposed_{b}\right) + \beta_{2}^{base} underreported_{ib} \\
& + \alpha_{2}\mathbf{X}_{ibt} + \epsilon_{ibt},
\end{aligned}
$$

where i, b, and t index firms, banks and quarters respectively.[23] The main explanatory variables are $exposed_b$, a dummy variable that is one for banks exposed to the EBA intervention and $underreport_{ib}$, a dummy that is one if the lending relationship has underreported loan losses in the four quarters prior to the announcement of the intervention. This dummy is based on our measure of underreporting described in Section I.

We use $period_\tau$ as a dummy that indexes periods of three quarters. The periods of interest are the EBA intervention (2011:IV–2012:II) and the period following the EBA deadline (and bank bailout) (2012:III–2013:I). We also include two pre-period dummies and one postbailout period dummy, all of which are of equal length.[24]

Additionally, φ_b is a bank fixed effect and $\mathbf{X}_{ibt}$ denotes relationship level controls.[25] Standard errors are two-way clustered at the firm and bank level.[26] We follow the literature and estimate the effect on changes rather than (log) levels. The growth rate of credit is our dependent variable: $g_{ibt}^{credit} = credit_{ibt}/credit_{ib,t-1} - 1$. The growth rate allows us to decompose the total change in credit into the portion coming from overdue credit and the portion coming from performing credit.[27] This decomposition is important to rule out that observed changes in total credit are driven solely by some firms paying down overdue credit and underreported firms accumulating more overdue credit.

The firm × quarter fixed effects, θ_{it}, control for the firm-level changes in credit growth and also absorb any time-variation induced for example by seasonality. This implies that we compare changes in the share of credit coming from exposed and nonexposed bank to the same firm (Khwaja and Mian 2008). This estimator requires firms to have multiple lending relationships, which is true for 56 percent of firms in our sample. We also run a model with separate firm and quarter fixed effects which then also includes firms that only have a single lending relationship.

The coefficients of interest are $\beta_\tau^{treatgroup}$ on the triple interaction, which estimate the treatment effects for the subset of underreported firms. Our hypothesis is that the EBA intervention increased distorted lending incentives for exposed banks. We therefore expect this coefficient to be positive during the EBA intervention. Given that the differential incentives disappear with the government bailout, we expect $\beta_\tau^{treatgroup}$ to either turn to zero (or negative) following the EBA deadline.

We also estimate the baseline treatment effects for all other firms, β_τ^{treat} for two reasons. First, the existing literature suggests that a tightening of capital requirements can lead banks to shed assets and decrease credit supply. We want to test whether the effect is present in this setting. Second, the total treatment effect for the subset of interest, firms with underreported losses, is $\beta_\tau^{treat} + \beta_\tau^{treatgroup}$.

[23] We condition on relationships that are present throughout the entire period of analysis. In a separate specification, we investigate the effect on the probability that a lending relationship is cut.

[24] The two pre-periods allow us to test for pre-trends in credit allocation, while the inclusion of the post-bailout period allows us to study the evolution of credit following the EBA deadline. The sample period includes 2009:I–2014:IV which allows us to estimate each β_τ. This implies that the quarters not contained in any of the period dummies are the omitted base group. A standard difference-in-differences would omit the t-2 and t-1 terms and include only a single post coefficient which would summarize the average treatment effect in the postperiod.

[25] The relationship controls are the lending share of the bank, the length of the relationship, a dummy if the bank is the main lender, the share of the firm in the bank's loan portfolio.

[26] We also run a version with standard errors only clustered at the bank level.

[27] Results, available upon request, are similar when using the log changes.

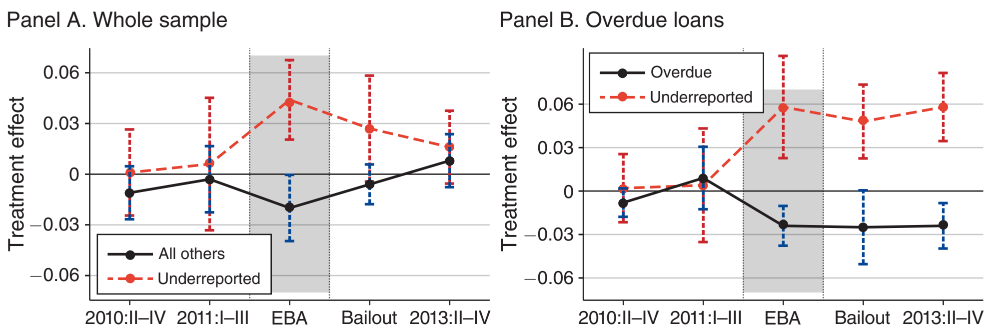

FIGURE 3. FIRM-BANK LEVEL CREDIT RESULTS

Notes: The graphs show results of the firm-bank level credit regression in specification (2), which includes firm $\times$ time and bank fixed effects as well as firm-bank-level controls ($N = 1{,}981{,}219$). The dependent variable is quarterly credit growth. We plot the coefficients on the two interactions $period_\tau \times exposed_b$ and $period_\tau \times underreported_{ib} \times exposed_b$, which are the respective treatment effects for the baseline group of firms (β_τ^{treat}) and the group of firms subject to loss underreporting ($\beta_\tau^{treatgroup}$). In panel B, we plot the coefficients from the same specification when conditioning on the sample of relationships with overdue loans ($N = 426{,}127$). Standard errors are clustered at the firm and bank level. The shaded areas mark the period of the EBA intervention. See Table A2 in the online Appendix for point estimates.

Results.—Figure 3 panel A shows our main credit results (see also Table A2 in the online Appendix for corresponding point estimates). Following the announcement of the EBA intervention, exposed banks increase credit supply to firms in financial distress that are subject to prior loss underreporting. The coefficient on the triple interaction of $period_\tau \times underreported_{ib} \times exposed_b$ in equation (2) is positive and strongly significant during the EBA intervention. This positive treatment effect for underreported distressed firms contrasts with the reduction in credit supply for all other lending relationships at exposed banks. The coefficient on $EBA_t \times exposed_b$ in equation (2) is negative and statistically significant (Figure 3 panel A and columns 2 and 3 of Table A2 in the online Appendix).

The magnitude of the shock is large. The baseline treatment effect of borrowing from exposed banks is a 2 percentage point drop in quarterly credit growth between the announcement and deadline of the EBA intervention. In contrast, the treatment effect for underreported firms is an increase in credit growth at exposed banks of just over 2 percentage points.[28] These changes are equivalent to 4 percent of a standard deviation of credit growth.

If loss underreporting correctly identifies firms which benefit from additional lending due to banks' distorted incentives, we should find that exposed banks do not increase credit supply to firms that are distressed but are not subject to underreporting. In Figure 3 panel B, we show results from running specification 2 in the sample of relationships that have overdue loans but are not subject to underreporting prior to the intervention. We find that the same reallocation patterns holds in this subsample holds, with non-underreported relationships experiencing a decline in credit supply and only underreported relationships experiencing an increase in credit.

[28]The total treatment effect adds the baseline treatment effect and the treatment effect for the subgroup of underreported firms.

The results suggest that the effects are driven by changes in bank credit supply in response to the EBA intervention. There is no evidence of differential credit allocation at exposed banks in the two periods prior to the intervention, lending credibility to our parallel trends assumption. The lack of pre-trends applies to both the baseline group of firms and to the subgroup of underreported firms. Second, the preferential credit treatment for underreported firms only occurs during the period of the EBA intervention when exposed banks face heightened distorted lending incentives. Similarly, the credit crunch only occurs in the period of the EBA intervention when banks attempt to comply with tighter capital requirements. While the differential treatment effect in growth rates disappears with the EBA deadline, the effect is persistent in levels. That is, we do not find evidence of *negative* treatment effects for underreported firms in the periods after the EBA intervention. This suggests that banks do not withdraw the additional credit granted during the EBA intervention following the EBA deadline. We provide a series of further robustness checks in Table A3 in the online Appendix.[29] To facilitate the interpretation of our results and ensure that our results are driven by a change in behavior of exposed banks, we also present results for difference-in-difference specification run separately for the sample of underreported and non-underreported firms (see Figure A5 in the online Appendix).

The results suggest that banks actively change their lending behavior during the EBA intervention. The change in total credit is almost entirely driven by performing credit (column 4 of Table A2 in the online Appendix). If underreported firms were simply converting more of their performing loan balances into overdue loans, we would expect no change in total credit, a reduction in performing credit, and an increase in overdue credit. Instead, we find an increase in total credit, an increase in performing credit, and a (statistically insignificant) reduction in overdue credit. Moreover, we find similar patterns when looking at the probability that a bank grants a new loan. We construct a dummy that is one if there is a new loan in a firm-bank relationship.[30] Column 6 of Table A2 in the online Appendix shows that we find a large significant increase in the probability that a new loan is granted to a underreported firms at exposed banks in the period of the EBA intervention. In contrast, the probability declines for all other firms at exposed banks.

The differential credit behavior is also visible at the extensive margin. The probability that an exposed bank cuts a relationship increases by almost 6 percentage

[29] We show that the estimated treatment effects are robust to the inclusion of firm-level controls averaged over the pre-period and interacted with period dummies. We also show that the estimated treatment effects are robust to differential clustering of standard errors, excluding relationship controls, and including the LTRO take-up.

[30] Our definition of a new loan requires that the total number of loans in a firm-bank relationship increases and that the total loan balance in the firm-bank relationship increases. While the credit register data do not allow us to track individual loans, banks report each individual lending operation to a given firm allowing us to count the number of loans in each period. Since existing loans can be split into several loans due to, for example, a restructuring operation we also impose the second condition on the total loan balance.

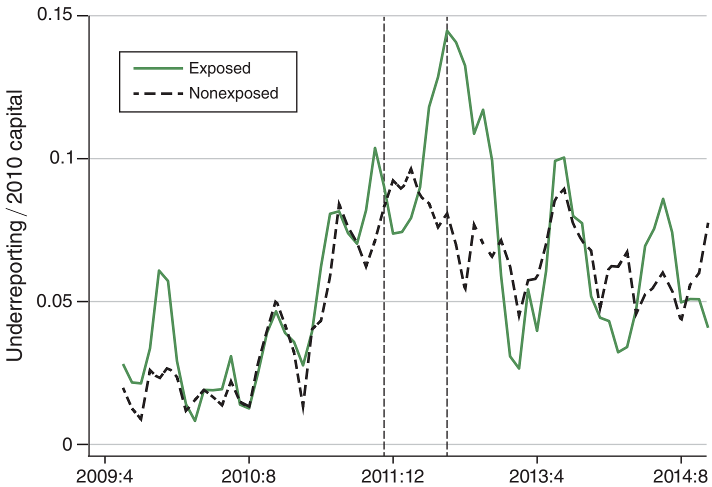

FIGURE 4. UNDERREPORTING OF LOAN LOSSES BY EXPOSED AND NONEXPOSED BANKS

Notes: This figure shows the evolution of aggregate underreported losses for exposed and nonexposed banks. Underreported losses are scaled by 2010 bank capital. The first vertical line denotes the announcement of the EBA intervention. The second vertical line denotes the EBA compliance deadline.

points during the EBA intervention (Table A4 in the online Appendix).[31] In contrast, the probability falls for underreported firms.[32]

Mechanism.—We find no evidence that the reallocation of credit to underreported firms is driven by risk-shifting behavior. Instead, our results are most consistent with banks delaying the recognition of losses.

First, Figure 4 shows that the extent of underreporting at the bank-level is very sensitive to the EBA intervention. The quantity of loans that are underreported sharply increases with the announcement of the EBA intervention. This increase lasts until the EBA deadline, at which point immediate regulatory capital pressures subside and exposed banks roll back the additional underreporting.

Second, we find no evidence of risk shifting when looking at risky lending to new or existing clients. Table 3 shows results from a series of difference-in-difference specifications where we regress various indicators of new lending on a bank exposure dummy as well as an interaction with firm risk. We focus on the sample of non-underreported firms. Our preferred measure of firm risk is predicted default from a credit risk prediction model based on Antunes, Gonçalves, and Prego (2016).

[31] Our indicator is a dummy that turns one in the month the performing credit balance drops to zero. We focus on the performing credit stock since banks often report relationships that only have nonperforming credit to the credit register for a very long time even when the credit is fully written off. The reason is that banks wait for the conclusion of the official insolvency process to stop reporting the debt to the credit register. Given very lengthy bankruptcy procedures in Portugal, this implies that nonperforming loan stocks can be reported in the credit register for years even though there no longer exists a meaningful credit relationship.

[32] We cannot estimate pre-trends in this specification since we condition on the sample of relationships with positive loan balances in the pre-period. Since we estimate the cumulative effect of existing a lending relationship, the dummy for exit remains 1 following the quarter of exit and contributes to the estimated probability in all subsequent quarter, the changes in the coefficients are informative about the additional exit. This implies that as in the intensive margin, the effects predominantly take place during the EBA intervention.

Table 3—Regression Results at the Firm-Bank Level: Risky Lending

Sample	New clients			Existing clients	
Outcome	log(new loans)		Approved	New loan	
	(1)	(2)	(3)	(4)	(5)
$Pre1_t \times exposed_b$	0.196	0.151	0.006	−0.014	−0.020
	[0.226]	[0.221]	[0.005]	[0.005]	[0.006]
$Pre2_t \times exposed_b$	0.278	0.262	0.000	−0.009	−0.007
	[0.280]	[0.223]	[0.007]	[0.006]	[0.006]
$EBA_t \times exposed_b$	−0.536	−0.548	0.002	−0.024	−0.022
	[0.302]	[0.250]	[0.004]	[0.008]	[0.009]
$Bailout_t \times exposed_b$	−0.213	−0.062	0.003	−0.009	−0.010
	[0.337]	[0.296]	[0.003]	[0.007]	[0.008]
$Postbailout_t \times exposed_b$	−0.009	−0.060	−0.001	0.005	0.005
	[0.295]	[0.261]	[0.004]	[0.003]	[0.003]
$Pre1_t \times exposed_b \times firm\ risk_i$	−0.887	0.043	−0.024	−0.004	0.007
	[0.572]	[0.056]	[0.017]	[0.003]	[0.003]
$Pre2_t \times exposed_b \times firm\ risk_i$	0.184	0.125	−0.035	0.003	0.001
	[0.999]	[0.053]	[0.018]	[0.005]	[0.003]
$EBA_t \times exposed_b \times firm\ risk_i$	−3.375	−0.157	−0.022	0.003	−0.001
	[1.274]	[0.053]	[0.016]	[0.004]	[0.004]
$Bailout_t \times exposed_b \times firm\ risk_i$	−1.722	−0.214	−0.017	−0.004	−0.005
	[1.269]	[0.129]	[0.012]	[0.003]	[0.005]
$Postbailout_t \times exposed_b \times firm\ risk_i$	−3.337	−0.073	−0.001	−0.002	−0.003
	[0.704]	[0.074]	[0.015]	[0.003]	[0.003]
Firm × quarter fixed effects	N	N	N	Y	Y
Firm quarter fixed effects	Y	Y	Y	N	Y
Relationship controls	N	N	N	Y	Y
Observations	138,781	165,801	6,373,525	1,941,026	1,918,000
R^2	0.116	0.111	0.010	0.413	0.378
Banks	45	45	45	45	45

Notes: The table shows regression results at the firm-bank level for new loans either to existing or new clients. The dependent variable is the logarithm of loans granted to new clients in columns 1 and 2, the approval rate of credit applications in column 3, and a dummy for whether an existing client receives a new loan in columns 4 and 5. Firm risk is measured by predicted default risk based on the credit risk prediction model of Antunes, Gonçalves, and Prego (2016) in columns 1, 3, and 4. In columns 2 and 5, firm risk is measured by cyclicality of firm sales. All regressions have bank fixed effects and cluster standard errors at the firm-bank level.

We find that new loans to both new and existing clients decline at exposed banks during the EBA intervention. Importantly, this decline is stronger for firms with higher predicted credit risk. This finding suggests that the credit reallocation to underreported firms is unlikely to be driven by risk-shifting incentives.

Credit Effects at the Firm-Level.—To detect whether firms undo effects at the firm-bank level by adjusting their credit coming from nonexposed banks, we analyze changes in credit allocation at the firm-level. We run the following dynamic differences-in-differences specification:

$$(3)\quad \Delta \log credit_{it} = \sum_{t=-5}^{10} \Delta_t^{treatgroup}\big(quarter_t \times treatment_i \times underreported_i\big) + \sum_{t=-5}^{10} \Delta_t^{treatment}\big(quarter_t \times treatment_i\big) + controls + \alpha_1 \mathbf{X}_{it} + \theta_i + \epsilon_{it},$$

where *treatment*$_i$ is the firm-level borrowing share from exposed banks prior to the intervention.[33] We standardize this variable to be able to interpret coefficients as the percentage change in credit in response to a standard deviation increase in the borrowing share from exposed banks. We use *underreported*$_i$ as a dummy that captures firms with underreported losses prior to the announcement of the intervention. Standard errors are clustered at the firm-level.

In contrast to the firm-bank level specification, we can no longer control for the firm-level change in total credit, which captures changes in credit demand. We therefore include a range of firm-level controls interacted with quarter dummies to allow for flexible differences in time trends across firms. These controls include two-digit industry and several firm characteristics averaged over 2008–2010 (sales growth, capital/assets, interest paid/EBITDA and the current ratio). The inclusion of controls accounts for potential long-term trends at the firm-level that could affect credit demand.

Results.—Figure 5 panel A shows our main credit results at the firm-level. Following the announcement of the EBA intervention, underreported firms with a larger borrowing share from exposed banks experience faster growth in credit than underreported firms who are less reliant on exposed banks. At the same time, credit declines for all other firms with a larger borrowing share from exposed banks. Both effects shift back to zero following the bank bailout at the EBA deadline. We hence confirm that the credit reallocation at the firm-bank level is also present at the firm-level, suggesting that firms cannot undo the effects at the firm-bank level.

Unlike in the firm-bank results, the positive treatment effect for underreported firms does not immediately revert after the bank bailout at the EBA deadline. This persistent effect on total credit is driven by an increase in overdue credit which begins after the EBA deadline (see Figure A8 in the online Appendix). This result suggests that banks can stave off additional default for underreported firms in the short-run but not in the medium to long-run. This result, together with the absence of pre-trends at the firm-level, provides further support for the argument that the credit reallocation is not driven by underlying differences in firm-level quality or liquidity trends. The increase in credit during the EBA intervention is again driven by performing credit as shown in Figure 5 panel B.

The economic significance of the credit reallocation is large. For underreported firms, the total treatment effect of borrowing exclusively from exposed banks versus borrowing exclusively from nonexposed banks is equal to a 16 percent increase in total credit relative to the base quarter (2011:III).[34] For all other firms, the total treatment effect is a decline in credit of 14 percent relative to the base quarter.

[33] Following Chodorow-Reich (2014), this variable is defined as $treat_i = \frac{\sum_{b=1}^{B^{exp}} L_{ib,pre}}{\sum_{b=1}^{B^{all}} L_{ib,pre}}$ where $L_{ib,pre}$ denotes the stock of total credit of firm i at bank b in 2010; B^{exp} is the set of exposed banks, while B^{all} is the set of exposed and nonexposed banks.

[34] This is the cumulative effect over the combined EBA and bailout period, which runs from 2011:III to 2013:I. A standard deviation in the borrowing share in our sample is the equivalent of moving from borrowing entirely from exposed to borrowing entirely from nonexposed. For underreported firms, this is the total treatment effect $\beta_\tau^{treat} + \beta_\tau^{treatgroup}$ in equation (3), or in other words, we add the two coefficients displayed in Figure 5 panel A.

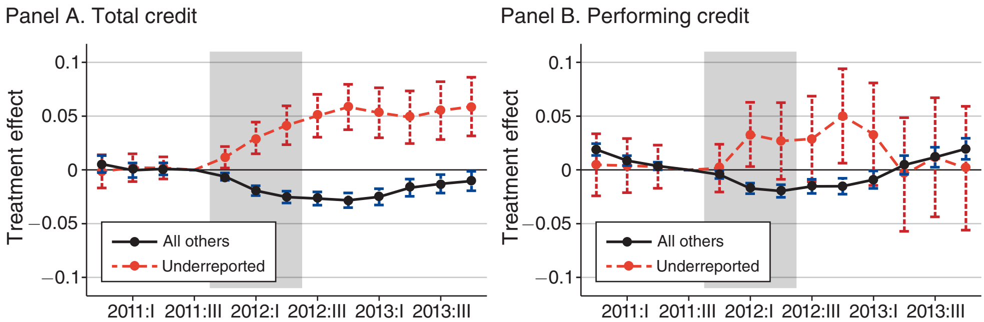

FIGURE 5. FIRM-LEVEL CREDIT TREATMENT EFFECTS

Notes: The graphs show results of the firm-level credit regression in specification (3). The dependent variable is the quarterly log of total credit for a given firm in panel A and the quarterly log of performing credit in panel B. We plot the coefficients on the two interactions $quarter_t \times treatment_i$ and $quarter_t \times treatment_i \times underreported_i$, which are the treatment effects for the baseline group of firms, and the group of firms subject to loss underreporting. The shaded areas mark the period of the EBA intervention. The specification includes the full set of interactions, industry $\times$ quarter and firm fixed effects, as well as firm-level controls interacted with quarter. All coefficients should be interpreted as changes in the dependent variable relative to the (normalized) base quarter 2011:III. Standard errors are clustered at the firm-level. $N = 1{,}346{,}771$.

Potential Threats to Identification.—The validity of our results rests on the assumption that the credit reallocation to underreported firms by exposed banks is not driven by an increase in credit demand by underreported firms. For this assumption to be violated in the context of our triple-difference design, banks have to underreport firms with better long-run fundamentals, those firms have to experience temporary financial distress driving up their credit needs coinciding exactly with the duration of the EBA intervention, and the nature of lending relationships has to be such that only exposed banks are in a position to respond to these additional credit needs.

To address this possibility, we first provide evidence that observable characteristics of underreported firms are not systematically correlated with how much they borrow from exposed banks prior to the EBA intervention (see Figure A6 in the online Appendix). Turning to the firm-bank level, Figure A6 in the online Appendix shows that EBA banks are no more likely to be the main lender, to grant a different level of credit, or to have a different share of performing credit. EBA banks seem to have slightly longer lending relationships and firms on average account for a larger share in the EBA banks' loan portfolio. These differences likely reflect that exposed banks on average are larger and have been present in Portugal for longer. To account for these differences, we control for relationship characteristics in all firm-bank level specifications. The fact that our lending result are robust to including measures of the strength of the firm-bank relationships also suggests that our results are unlikely to be driven by exposed banks protecting their most loyal borrowers from cuts in credit supply.

Second, we investigate the potential presence of differential financial shocks driving observed outcomes. Given that we absorb any firm-level changes by firm $\times$ time fixed effects in our main specification, differential shocks to credit demand provide a potential challenge only for our firm-level regressions. At the firm-level, the main difficulty for confounding firm-level financial shocks to explain the results stems from the fact that the EBA intervention is temporary. For concurrent liquidity shocks to explain the results, we would need that firms borrowing from exposed banks experience a negative liquidity shock, leading to a positive credit demand shock, at the time of the EBA intervention and that this shock dissipates with the onset of the EBA deadline. Nonetheless, we provide evidence against different liquidity trends prior to the shock by estimating a dynamic firm-level difference-in-difference regression with liquidity ratios as the dependent variable. Figure A7 in the online Appendix shows that there are no pre-trends in either the current ratio or the cash/assets ratio for these firms.

One remaining potential issue is that underreported firms may be aware of their special status and also aware of the EBA intervention affecting their lender. Firms could use the intervention to extract additional credit from the bank by threatening immediate default on outstanding payments, which would impose a loss on the bank at a time when bank capital is scarce. According on anecdotal evidence, firms are passive actors in banks' reporting management and likely unaware of whether or not they are underreported. However, even if this mechanism were in operation, it would be consistent with the distorted lending channel that this paper documents.

III. Measuring the Effect on Aggregate Productivity Growth

We show that the EBA-induced credit reallocation to underreported firms contributes negatively to aggregate productivity growth. First, we show that the marginal output gain from allocating additional inputs to underreported firms is much smaller relative to nondistressed firms in the economy, based on the popular approach of measuring wedges in firms' first-order conditions (Restuccia and Rogerson 2008; Hsieh and Klenow 2009). Second, we show that the credit reallocation to underreported firms leads to an increase in the dispersion of wedges across firms. Finally, using a decomposition of aggregate productivity growth, we show that this widening dispersion of distortions can account for about 13 percent of the decline in productivity in Portugal in 2012.

A. *Measuring Firm-Level Distortions*

When exposed banks reallocate credit from nondistressed firms to underreported firms, they prevent inputs held by underreported firms from being reallocated to firms where these inputs would have potentially earned higher returns. At the same time, credit taken up by underreported firms shrinks the available credit supply for firms with potentially high factor returns, forcing them to shed resources. Intuitively, unconstrained firms hire labor and invest in capital up to the point where marginal revenue products equal their user costs. However, distortions, such as financial frictions, taxes, monopoly power or market failures, prevent firms from hiring sufficient inputs to drive down the marginal revenue products to the first-best level and lead to positive wedges in the first-order conditions for the first-best resource allocation. In an economy with such distortions, there are productivity gains from allocating resources away from firms that are close, or even beyond, their optimal factor use (small wedges) towards firms which are far away from their optimal factor use (high wedges).[35]

[35] Technically, we require dispersion in wedges for allocative inefficiency to arise. If all firms face the same distortions, there is no reason to redistribute inputs.

Table 4—Capital and Labor Wedges

		Nonperforming	
	Performing	Underreported	Not underreported
Panel A. Capital and labor wedges			
MRPL	48.24	37.30	42.39
	(39.86)	(36.37)	(37.73)
w	12.23	10.34	11.21
	(7.66)	(6.84)	(7.61)
Labor wedge	4.24	3.75	4.03
	(3.81)	(3.23)	(3.19)
MRPK	63.16	47.44	55.51
	(38.14)	(40.22)	(39.35)
$r + \delta$	14.70	13.33	15.02
	(12.01)	(11.35)	(12.02)
r	5.12	5.95	6.61
	(6.59)	(6.92)	(7.65)
δ	10.31	7.79	8.90
	(10.33)	(8.46)	(9.12)
Capital wedge	5.23	4.10	4.49
	(3.88)	(3.79)	(3.71)
Observations	106,680	4,553	15,362
	Labor	Capital	
	(1)	(2)	
Panel B. Persistence of wedges			
$\beta wedge_{t-1}$	0.786	0.594	
	(0.002)	(0.002)	

Notes: This table shows means and standard deviation for the inputs and capital and labor wedges. Panel A shows summary statistics for labor and capital wedges by type of firm in 2009–2011. Performing loans refers to firms with only performing loans. Nonperforming refers to firms with some nonperforming loans. Underreported refers to firms with underreported losses. *MRPL* (*MRPK*) refers to the marginal revenue product of labor (capital). Interest rates (r), depreciation rates (δ), and *MRPK* are expressed in percent. Wage (w), labor gap, and *MRPL* are expressed in thousands of euros. Wedges are measured using firm-level price. Panel B reports coefficients from regressing capital and labor wedges on the lagged wedge to gauge the extent of persistence.

Firm-level wedges for labor and capital in a Cobb-Douglas framework are defined as

$$\alpha_s \frac{Y_{it}}{K_{it}} = (r_t + \Delta_t)(1 + \tau_{it}^K), \tag{4}$$

$$\beta_s \frac{Y_{it}}{L_{it}} = w_t(1 + \tau_{it}^L); \tag{5}$$

where Y_{it} is value added, L_{it} is the number of employees, and K_{it} is the capital stock, which we compute using the perpetual inventory method. See Appendix C on details for the underlying theoretical framework and details on how we compute the wedges in the data.

Table 4 compares marginal products of labor and capital and the resulting wedges at underreported firms and non-underreported firms. Underreported firms have substantially lower marginal revenue products relative to nondistressed firms.

These numbers suggest that reallocating a worker from an underreported firm to a nondistressed firm would on average increase output by €10,940 (the difference between a marginal revenue product of labor of 48,240 at non-underreported firms and 37,300 at underreported firms). For capital, the difference in marginal products imply that a unit of capital of capital would yield about a 16 percent higher return when reallocated away from a non-underreported firm to a performing firm. When comparing firms with nonperforming, but correctly reported, loans (distressed but not underreported) to the underreported firms, the differences in marginal products and wedges largely disappear. This finding implies that, conditional on distress, underreporting is not systemically correlated with marginal productivity.

We address two potential challenges in the measurement of the wedges. First, there is a potential worry that the smaller marginal products for underreported firms reflect temporary shocks and that long-run productivity for these firms is in fact higher. Panel B in Table 4 however shows the firm-level wedges are very persistent, with an annual correlation of 60–80 percent. This is also consistent with the evidence from Figure A4 in the online Appendix which shows that underreported firms do not show positive reversals in return on assets, sales growth, or the share of nonperforming loans in the long-run.

Second, there is a concern that low marginal products of capital for underreported firms may simply reflect that these firms are less risky, which is reflected in a lower return on capital (David, Schmid, and Zeke 2022). In Table A6 in the online Appendix, we show that the marginal revenue product of capital for underreported firms remains significantly lower even when controlling for measures of firm-level risk, such as cyclicality of sales, volatility of firm sales, and predicted default risk.[36] These results also address concerns that wedges may more generally be the result of adjustment costs, time-varying markups, or volatility in firm-level total factor productivity (TFP) (Asker, Collard-Wexler, and DeLoecker 2014). Table A6 in the online Appendix shows that our measurement of marginal products is robust to controlling for firm-level volatility of sales which capture some of these alternative explanations. Moreover, we show in the next section that wedges respond to firm-level variation in credit supply shocks, suggesting that wedges are, at least partly, driven by credit frictions.

B. *Effect on Inputs and Wedges*

We use the following instrumental variable design to estimate the effect of the EBA-induced credit reallocation on input use and input wedges:

$$\Delta \log y_{is} = \gamma \Delta \log credit_{is} + controls + u_{is}, \tag{6}$$

where i and s index firms and industries, respectively.

[36] Since most firms in our sample are private, we cannot compute stock market based risk factors used in David, Schmid, and Zeke (2022).

We instrument for $\Delta \log credit_{is}$ with the nondynamic version of the firm-level credit regression in equation (3).[37] The dependent variables are the change in labor and capital and the growth rate of capital and labor wedges.[38]

For labor and capital, we use the Davis-Haltiwanger symmetric growth rate (Davis, Haltiwanger, and Schuh 1996), which is a second-order approximation to the log difference growth rate that takes into account observations that turn to zero.[39] The change in wedges is estimated in the sample of continuing firms. We estimate this regression for 2012 since the firm-level data are only available at an annual frequency and 2012 is when most of the EBA intervention occurs. To address concerns that treated firms may have been on different long-term trends, we include a lag of the dependent variable.

The instrumental variable (IV) results in panel B of Table 5 suggest that the credit reallocation leads to a reallocation of labor and capital to underreported firms. The elasticity of labor with respect to a change in credit supply is 0.52 while the elasticity with respect to capital is 0.14. These numbers combine the intensive and extensive margin response (firm exit).

The average underreported firm has a 40 percent borrowing share from EBA banks which, based on the first stage, implies a 13 percent average increase in credit. Multiplying this by the second-stage coefficient, we get that labor and capital use at underreported firms *increase* on average by 6 percent (0.47×0.13) and 9 percent (0.68×0.13). For nondistressed firms, the first stage estimates imply an average decline in credit of 12 percent, implying that labor and capital *decline* by 6 percent (0.47×-0.12) and 8 percent (0.68×-0.12), respectively. Distressed but correctly reported firms experience an average decline in credit of 9 percent (due to a smaller borrowing shares from EBA banks), implying a decline in labor and capital of 4 percent (0.47×-0.09) and 6 percent (0.68×-0.09), respectively. The first-stage F-statistics are all comfortably above the Stock and Yogo (2005) criterion for 5 percent maximal bias.

Specification (6) assumes that elasticities are homogeneous across all firms. For the aggregation exercise in the next section, we estimate a specification that measures the effect on wedges (which depend on both changes in output and inputs) and allows the elasticity to depend on the degree of financial constraints faced by the firm. We measure financial constraints by the pre-EBA level of the capital wedge which reflects financing frictions that prevent firms from hiring the optimal amount of capital. Panel A of Table 5 shows that the effect of the EBA intervention intensify with the level of financial constraints.

The real effects of the EBA intervention dissipate post-2012 (see Table A5 in the online Appendix). However, it is difficult to precisely estimate the long-run effects since the credit intervention is short-lived and hence the instrument loses power after

[37] The first-stage equation is

$$\Delta \log credit_{is} = \Delta^{treatment} borrowing\ share_{is} + \Delta^{treatgroup} borrowing\ share_{is} \times underreported_{is} + controls + \epsilon_{is}.$$

[38] Since allocative efficiency depends only the relative marginal productivities, we use the growth rate of the marginal product of labor and capital as our main explanatory variable that will feed into the decomposition in the following section.

[39] The formula is $g^y = \frac{y_t - y_{t-1}}{0.5(y_t + y_{t-1})}$. This is a popular way of defining the outcome variable in labor and increasingly corporate finance. The symmetric growth rate is bounded between -2 and 2.

Table 5—Effects on Wedges, Inputs, and TFP

	Labor		Capital		
	OLS	IV	OLS	IV	
	(1)	(2)	(3)	(4)	
Panel A. Wedges					
Share EBA_i	0.007		0.017		
	[0.004]		[0.008]		
Share EBA_i × *'11 capital wedge*	0.012		0.028		
	[0.008]		[0.013]		
$\Delta \log credit_i$		−0.008		−0.499	
		[0.064]		[0.159]	
$\Delta \log credit_i$ × *'11 capital wedge*		−0.130		−0.569	
		[0.091]		[0.202]	
	Labor		Capital		TFP
	OLS	IV	OLS	IV	IV
	(1)	(2)	(3)	(4)	(5)
Panel B. Labor, Capital, TFP					
$\Delta \log credit_i$	0.050	0.522	0.105	0.140	0.001
	[0.002]	[0.094]	[0.004]	[0.046]	[0.005]
First-stage *F*-statistic		111.2		111.2	111.2
Observations	104,499	104,499	104,499	104,499	104,492

Notes: The table shows IV regression results at the firm level for 2012. The dependent variables are symmetric growth rates, which are second order approximation to the log difference growth rate. Capital refers to fixed assets. Labor refers to the number of employees. We instrument for the log change in credit using the (normalized) firm-level borrowing share from banks exposed to the EBA shock prior to the shock interacted with the underreported firm dummy. Controls consist of firm-size and two-digit industry fixed effects, as well as firm-level log total assets, interest/EBITDA, capital/assets, current ratio, cash/assets and sales growth, all averaged over 2008–2010. Capital and labor wedge are defined in the text. The '11 capital wedge refers to the 2011 firm-level capital wedge, which is used as an indicator of financial constraints. The dependent variable in panel B column 56) is firm-level TFP. Standard errors are clustered by industry, in brackets. OLS = ordinary least squares.

2012. We also conduct placebo exercises running the same specification in the years prior to the intervention and find no significant effects (see Table A5 in the online Appendix).

C. *Decomposing Productivity Growth*

We use a decomposition of productivity growth to aggregate our estimates of firm-level changes due to the EBA-induced credit reallocation to underreported firms. The advantage of this approach that it allows us to aggregate firm-level estimates without having to assume a specific model of the economy. The disadvantage is that it will not allow us to take into account potential general equilibrium effects.

We use a decomposition following Osotimehin (2019) which expresses aggregate productivity growth as the sum of the change in technical efficiency (TE) and changes in allocative efficiency (AE) both within and between sectors:[40]

$$\Delta \ln TFP \simeq \Delta TE + \Delta AE_{within} + \Delta AE_{between}. \tag{7}$$

[40] We abstract from the exit-entry component since Osotimehin (2019) shows that this component is small. Instead, we work with the sample of continuing firms.

The total change in technical efficiency describes the change in firm-level productivity for a given input allocation and is given by the following weighted sum of technical efficiency across sectors

$$\Delta TE = \sum_{s=1}^{S} \frac{1}{1-\gamma_s \rho}\left[s_t^Y - \rho \sum_x^X \epsilon_t^x s_t^x\right] \Delta TE_s, \tag{8}$$

where s_t^x is the sectoral share of input x. For example, the sectoral capital share is K_{st}/K_t. In our setting, the inputs are capital and labor (with $x = K, L$). $s_t^Y = \frac{P_{st}Y_{st}}{P_t Y_t}$ is the sectoral value added share; γ_s is the returns to scale parameter, ρ governs the elasticity of substitution across sectors, and ϵ^X is an elasticity defined in Appendix C. The change in sectoral technical efficiency, ΔTE_s is a function of firm-level TFP changes (see Appendix C for details).

The next two components capture changes in allocative efficiency of inputs across firms within and across sectors. In the absence of frictions, wedges are constant and aggregate TFP is only driven by changes in technical efficiency. However, in the presence of frictions and distortions, aggregate productivity growth also depends on changes in allocative efficiency. Allocative efficiency is measured based on the wedges between firms' marginal productivity similar to the approach in Hsieh and Klenow (2009). The within-sector component is a weighted sum of the allocative efficiency change in each sector:

$$\Delta AE_{within} = \sum_{s=1}^{S} \frac{1}{1-\gamma_s \rho}\left[s_t^Y - \rho \sum_x^X \epsilon_t^x s_t^x\right] \Delta AE_{within,s}. \tag{9}$$

The change in allocative efficiency in a given sector in turn is given by a weighted average of firm-level wedges:

$$\begin{aligned} \Delta AE_{within,s} = {} & \frac{\alpha_s}{1-\gamma_s\theta_s}\sum_i \left[(1-\beta_s\theta_s)s_{i,t-1}^K + \beta_s\theta_s s_{i,t-1}^L - s_{i,t-1}^Y\right]\frac{\Delta\tau_{it}^K}{1+\tau_{i,t-1}^K} \\ & + \frac{\beta_s}{1-\gamma_s\theta_s}\sum_i \left[\alpha_s\theta_s s_{i,t-1}^K + (1-\alpha_s\theta_s)s_{i,t-1}^L - s_{i,t-1}^Y\right] \\ & \times \frac{\Delta\tau_{it}^L}{1+\tau_{i,t-1}^L}. \end{aligned} \tag{10}$$

This expression implies that allocative efficiency deteriorates if the dispersion of wedges across firms in a sector grows. Since allocative efficiency depends only the relative marginal productivities, we use the growth rate of the marginal product of labor and capital in the empirical implementation.

Finally, the change in between-sector allocative efficiency is given by a weighted average of sectoral wedges:

$$\begin{aligned} \Delta AE_{between} = {} & \sum_{s=1}^{S}\frac{1}{1-\gamma_s\rho}\left[\epsilon_t^K(1-\beta_s\rho)s_t^K + \epsilon_t^L\alpha_s\rho s_t^L - \alpha_s s_t^Y\right]\frac{\Delta\mathcal{T}_{st}^K}{1+\mathcal{T}_{s,t-1}^K} \\ & + \sum_{s=1}^{S}\frac{1}{1-\gamma_s\rho}\left[\epsilon_t^K\beta_s\rho s_t^K + \epsilon_t^L(1-\alpha_s\rho)s_t^L - \beta_s s_t^Y\right] \\ & \times \frac{\Delta\mathcal{T}_{st}^L}{1+\mathcal{T}_{s,t-1}^L}. \end{aligned} \tag{11}$$

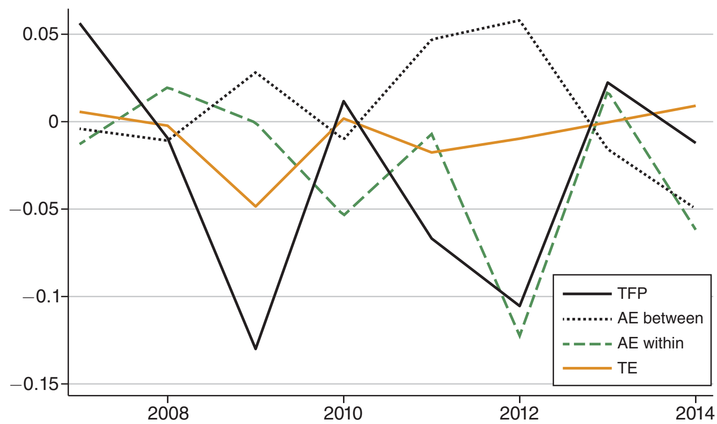

FIGURE 6. DECOMPOSITION OF AGGREGATE PRODUCTIVITY

Notes: This figure shows changes in aggregate TFP and the decomposition of these changes according to equation (7). "AE within" refers to changes in allocative efficiency within sectors while "AE between" refers to changes in allocative efficiency between sectors. TE refers to technical efficiency. The decomposition is an approximation and hence the components do not exactly add up to the total TFP changes.

Figure 6 applies the decomposition in equation (7) to firm-level data in Portugal. See Appendix C on details how each component is constructed. Figure 6 shows that Portugal, like other eurozone periphery countries, for the most part experienced negative productivity growth in the years around the sovereign debt crisis. This negative productivity growth is driven to a large extent by the deterioration of within-sector allocative efficiency.

D. *The Effect of the EBA Intervention*

We use the productivity decomposition in equation (7) to quantify the contribution of the EBA intervention to the decline in allocative efficiency and aggregate productivity in 2012. We first report the total effect of the EBA intervention, which we obtain by aggregating the firm-level changes in wedges estimated in Section IIIB. We then isolate the effect that the credit reallocation to underreported firms has on allocative efficiency and productivity. The former exercise asks what would have changed if the EBA intervention had not happened and no reduction in credit supply and no credit reallocation had occurred. The latter exercise asks what would have changed if the EBA intervention had been constructed to avoid the reallocation of credit to underreported firms, but still led to a contraction of credit supply. Both of these exercises are partial equilibrium.

The productivity decomposition shows that changes in productivity could in principle be driven by technical efficiency, between-sector allocative efficiency, or within-sector allocative efficiency. We focus on within-sector allocative efficiency since the contribution of between-sector allocative efficiency to total productivity is small (see Figure 6). Since we cannot reject the null of no effect on firm-level TFP (see Table 5 panel B) and there is no strong theoretical prior for expecting credit

Table 6—Effect of EBA Intervention on Allocative Efficiency and Productivity

		Of which:	
	Δ Allocative efficiency	Capital	Labor
	(1)	(2)	(3)
Panel A. Total effect of EBA intervention			
Estimated effect (percent)	−6.59	−4.15	−2.44
Percent of actual	53.83	45.78	35.38
Memo: actual (percent)	*−12.24*	*−5.34*	*−6.91*
Panel B. Effect of EBA credit reallocation			
Estimated effect (percent)			
Min	−0.89	−0.48	−0.40
Mean	−2.71	−1.47	−1.23
Max	−4.43	−2.33	−2.10
	Percent of actual		
Mean	22.10	23.09	17.85

Notes: This table shows the estimated contribution of the EBA intervention to the decline in allocative efficiency in 2012. Panel A shows the the total partial equilibrium effect summing over all firm-level changes in wedges. Panel B shows the effect of changes resulting from the reallocation of credit to underreported firms only. Min, mean, max refer to the results of the simulation described in Section IIID. Actual refers to the total decline in allocative efficiency in 2012 and is obtained from equation (7). Both exercises are explained in detail in Section IIID.

frictions to affect TFP, we exclude the potential TFP effects from our aggregation exercise.[41] In other words, we set the ΔTE term in equation (7) to zero.

To obtain the total partial equilibrium effect of the EBA intervention, we first compute predicted changes in labor and capital wedges for each firm by

$$(12) \qquad \frac{\Delta \hat{\tau}_i^X}{1 + \tau_{it,t-1}^X} = \left(\hat{\gamma}_1^X \; + \hat{\gamma}_2^X \times \textit{capital wedge}_{i,t-1}\right) \times \underbrace{\left(\hat{\Delta}^{treatment} \textit{borrowing share from EBA banks}_i\right)}_{\widehat{\Delta \log credit_i}}$$

for $X = K, L$. We multiply each firm's borrowing share from EBA banks by our first-stage coefficient to obtain the predicted change in credit supply for each firm. We then use the second-stage estimates in panel A of Table 5 to compute the predicted change in labor and capital wedges. As discussed in Section IIIB, we allow the effect on labor and capital wedges to vary with the ex ante level of financial constraints at the firm-level (as measured by the size of the lagged capital distortion: $\textit{capital wedge}_{i,t-1}$).

We then aggregate over these firm-level changes using equations (9) and (10). Table 6 shows that the EBA-induced changes suggest a decline in allocative efficiency of 6.6 percent, which represents about 54 percent of the total decline in within-sector allocative efficiency in Portugal in 2012.

[41] TFP residuals are not generally informative about firm-level wedges since there is no inherent reason to expect firm-level TFP residuals and distortions to be correlated (Restuccia and Rogerson 2008; Hsieh and Klenow 2017; Nishida et al. 2017).

We then isolate the effect of the credit reallocation to underreported firms by summing only over the subset of firm-level changes induced by the credit reallocation to underreported firms. This subset consists of all changes at underreported firms plus the changes at a randomly selected subset of non-underreported firms such that the total credit allocated to underreported firms matches the total decline in credit at the randomly selected subset of non-underreported firms. The assumption here is that the reduction in credit supply at the bank-level is dictated by the combination of existing bank equity and the shortfall to the EBA regulatory requirements. Hence if exposed banks allocate more credit to underreported firms, they must cut the equivalent amount of credit for some non-underreported firms. Since we do not know which firms the bank would cut credit for first, we draw 10,000 different samples of non-underreported firms. We then again use equations (9) and (10) to aggregate the predicted changes for the set of underreported firms plus the randomly selected sample.

Panel B of Table 6 shows that the credit reallocation to underreported firms on average reduces within-sector allocative efficiency by 2.7 percent which represents 22 percent of the decline in allocative efficiency in Portugal in 2012. The simulation also shows that the aggregate effect of the reallocation-only component can vary substantially. At one extreme, the credit reallocation can account for almost 36 percent of the drop in allocative efficiency. This case happens if banks withdraw credit from firms with higher-than-average wedges first, generating the largest dispersion of distortions and leading to the largest adverse effects on allocative efficiency. In contrast, banks might withdraw credit in a way that does not substantially drive up the dispersion of wedges. As shown by the minimum estimate in Table 6, withdrawing credit in this way implies that the reallocation to underreported firms accounts for only about 7 percent decline in allocative efficiency.

IV. Conclusion

This paper studies how distressed banks adjust to shortfalls in their ratio-based capital requirements during an economic downturn. We exploit a quasi-natural experiment in Europe that increased minimum regulatory capital requirements for a subset of banks. We show that affected banks further distort their reporting and lending choices and that this heightened credit misallocation by distressed banks lowers the allocative efficiency of capital and labor, adversely affecting aggregate productivity growth.

Our results suggest important policy conclusions. First, prompt corrective action by financial regulators is key to avoiding a situation in which regulators face a banking system which is already in distress. Good regulation is harder to design when banks are already weak. Instead, our results suggest that curbing leverage pre-crisis can avoid the risk that regulatory action heightens the type of distorted incentives of already distressed banks that we document in this paper. Second, our results highlight downsides of imposing capital regulation in the form of minimum required ratios of regulatory capital to risk-weighted assets. Giving banks a choice over whether to adjust the numerator or denominator opens up the possibility that banks distort both quantity and composition of credit supply to comply with the required ratios. Our paper provides a rationale to look beyond ratio-based requirements. It is important to stress that our results pertain to adjustments to ratio-based capital requirements in distressed conditions and do not apply to prudently increasing capital requirements when banks are not in distress.

Beyond regulatory policy, this paper highlights the importance of understanding the effect of credit supply frictions on the composition of credit, especially in economies where banks play a key role in allocating resources to firms. We show that in such bank-dependent economies, the (mis)allocation of credit feeds through to the (mis)allocation of production factors. The fact that credit supply distortions have an effect on the allocative efficiency of production factors is likely symptomatic of financial frictions present beyond the specific regulatory intervention we study. Our results therefore highlight a more general mechanism through which distressed banks impose economic costs.

Appendix A. A Method to Detect the Underreporting of Loan Losses

A1. *Notation*

We denote the observed loan balance reported in overdue bucket k in month t by $B_{ib}(t;k)$ where i denotes the firm and b the bank. We will drop the firm-bank subscripts in the discussion that follows. There are 14 reporting buckets of overdue which correspond to the overdue buckets in the regulatory schedule

$$k \in \big\{\{0\},\{1\},\{2\},\{3,4,5\},\ldots,\{30,\ldots,35\}\big\},$$

where 0 refers to loans overdue less than 30 days. We denote the set of available reporting buckets by K. The first three buckets are monthly, thereafter we observe three-month buckets and thereafter six-month buckets.

We also define a series of unobserved buckets c, which are defined at the monthly frequency $c \in \big\{\{0\},\{1\},\{2\},\ldots,\{35\}\big\}$. We also define an unobserved amount of lending $C(t;c)$, which is the loan balance in each of the unobserved monthly buckets. These underlying unobserved loan balances have to add up the observed distribution: $B(t;k) = \sum_{c\in k} C(t;c)$. We will exploit the fact that we can observe the first three monthly buckets in the data; that is, we can observe $C(t;c)$ for $c \in \{0,1,2\}$.

We first assume that there are no inflows or outflows, with the exception of entry mass $IN(t;0) = C_j(t;0)$ that enters the system in the lowest reporting bucket. We relax this assumption in the following section. In the absence of any inflows and outflows, it must hold that $C(t;c) = C(t-1;c-1)$.

Intuitively, the loan balance we observe in bucket c at t must be the loan balance that has moved up from the preceding bucket in the previous period. We define excess mass as the deviation from this identity:

$$E(t;b) = C(t;c) - C(t-1;c-1). \tag{A1}$$

A2. *Baseline Algorithm*

The goal of the algorithm is to compute the amount of excess mass in each reported overdue bucket k in each month t for each lending relationship.

We define the auxiliary concept of cumulative excess mass as

$$\bar{E}(t;k) = \sum_{j=1}^{s} E(t-j;k). \tag{A2}$$

Cumulative excess mass is the excess mass accumulated in a bucket k over the past s months where s denotes the length of the bucket (e.g., three months).

We proceed in two steps: we first calculate cumulative excess mass $\bar{E}(t;k)$ from the observed mass $B(t;k)$, and then recursively calculate excess mass $E(t;k)$ from the cumulative excess mass.

The algorithm consists of the following steps:

1) Set $E(-1;k) = 0 = E(0;k)$ for all $k \in K$.

2) For all $t = 1,\ldots,T$:

a. $E(t;\{1\}) = B(t;\{1\}) - B(t-1;\{0\})$.

b. $E(t;\{2\}) = B(t;\{2\}) - B(t-1;\{1\})$.

c. $\bar{E}(t;\{0,1,2\}) = \sum_{\tau=t-2}^{t}\left[E(t;\{1\}) + E(t;\{2\})\right]$.

d. For all $k = 4,\ldots,8$:

i. Cumulative excess mass

$$\bar{E}(t;k) = B(t;k) - B(t-3;k-1) + \bar{E}(t;k-1).$$

ii. Excess mass

$$E(t;k) = \bar{E}(t;k) - E(t-2;k) - E(t-1;k).$$

e. For $k = 9$:

i. $$\bar{E}(t,k) = B(t;k) - B(t-6;k-1) - B(t-6;k-2) + \bar{E}(t,k-1) + \bar{E}(t-3,k-1) + \bar{E}(t-3,k-2).$$

f. For $k = 10,\ldots,K$:

i. $\bar{E}(t,b) = B(t;k) - B(t-6;k-1) + \bar{E}(t,k-1)$.

ii. $E(t;k) = \bar{E}(t;k) - \sum_{\tau=1}^{5}\bar{E}(\tau;k)$.

We initialize the level of excess mass at zero in the month when our data is first available (January 2009) (step 1). For the first three buckets, we observe each month reported separately and hence directly use the baseline formula to calculate excess mass for the first two buckets (steps 2a–b). In step 2c, we obtain cumulative excess mass for the first combined three-month bucket at t by adding the excess mass in the first two buckets across the past three months. For the following buckets, we have to take into account that reporting is done in buckets that stretch over three or six months respectively. We first compute cumulative excess mass in step 2di by exploiting that the amount we observe in bucket k at time t is the sum of the amount that has been moved from the preceding bucket $k - 1$ over the course of the last three months minus the cumulative excess mass in the preceding bucket that was not transferred over the last three months, plus the cumulative excess mass that has stayed behind in bucket k over the past three months. Once we have calculated excess mass in step 2di, we can then recursively compute excess mass in step 2dii. We repeat similar steps for the six-month buckets.

A3. *Algorithm with Flows*

We now explain how to adjust the baseline algorithm for flows. We allow for the observed lending at t to be affected by time t inflows and outflows. The observed lending stock evolves as follows $stock_t = stock_{t-1} + net\ inflow_t$. We can further decompose net inflows into the following components:

$$net\ inflow_t = entry_t + installments_t - written\text{-}off_t - restructured_t + residual_t.$$

Inflows, other than the initial entry inflow, consist of installments that fall overdue. Inflows into buckets higher than the initial $k = 0$ will lead us to *overestimate* excess mass since these flows add to the observed mass at t. Since installments tend to be of fixed size and occur at regular intervals, we classify an increase in the overdue loan balance that corresponds to an exact decrease in the balance of performing credit and that occurs at least twice as an installment.

Outflows in contrast will lead us to *underestimate* excess mass since we subtract too much past mass. In the extreme case, this will lead us to obtain negative excess mass. Outflows happen for three reasons: repayment, restructuring, and write-offs. If a bank restructures or writes off an overdue loan, it reduces the overdue balance and increases the restructured/write-off balance which are separate entries in our data. We can therefore measure outflows into these two categories by a reduction in the overdue balance in a given bucket that is less or equal to the change in restructured/written-off balance in the same month. We cannot directly measure repayments of overdue loans which will instead be recorded as a (negative) residual. We distribute the residual across buckets by assigning the residual to the buckets with nonzero overdue balances in line with the share of lending reported in that bucket.

Since this distribution of residual flows to buckets is somewhat arbitrary, we conduct robustness checks to see how much the results change when shifting the residual flows to the lowest (highest) buckets. Since residual flows are small, the results are not affected by this assumption (see Figure A1 in the online Appendix).

The basic formula adjusted for flows is

$$\text{(A3)} \qquad E(t;k) = \left[C(t;c) - IN(t;c)\right] - \left[C(t-1;c-1) - OUT(t;c-1)\right].$$

We subtract inflows out of bucket c since these flows contribute to *observed* mass but do not contribute to *excess* mass. We add outflows from the preceding bucket since we do not expect these outflows to have moved up into the next reporting bucket. If we observed only monthly buckets, then we could again apply the simple formula to all buckets. However, for the three-month and six-month buckets, we again need to resort to the auxiliary concept of cumulative excess mass.

The formula for cumulative excess mass adjusted for flows is as follows:

$$\begin{aligned}\bar{E}(t,k) \ &= \ B(t;k) - B(t-3;k-1) + \bar{E}(t,k-1) \\ &\quad + \hat{OUT}(t;k-1) - \tilde{IN}(t;k-1) - \hat{IN}(t;k) + \hat{OUT}(t;k).\end{aligned}$$

The flow adjustments consists of the following components. We denote individual monthly buckets within each three-month bucket as $k\{1\}, k\{2\}, k\{3\}$. Hence $k\{2\}$ refers to the middle bucket within the three month bucket k.

1) $\hat{OUT}(t;k-1)$: For outflows, we want to subtract all outflows out of the preceding bucket over the past three months, which we would not have expected to have turned up in the current bucket. Specifically, these are the outflows from the "boundary" bucket $\{3\}$ that we would not expect to move across into the next bucket:

$$\begin{aligned}\hat{OUT}(t;k-1) \ &= \ OUT(t,(k-1)\{3\}) + OUT(t-1,(k-1)\{3\}) \\ &\quad + OUT(t-2;(k-1)\{3\}).\end{aligned}$$

2) $\tilde{IN}(t;k-1)$: There are inflows into the previous bucket $k-1$, some of which we expect to have moved by time t and we need to add:

$$\begin{aligned}&\tilde{IN}(t;k-1) \\ &= \underbrace{IN(t-3;(k-1)\{1\}) + IN(t-3;(k-1)\{2\}) + IN(t-3;(k-1)\{3\})}_{\text{already incorporated in } B(t-3;k-1)} \\ &\quad + IN(t-2;(k-1)\{2\}) + IN(t-2;(k-1)\{3\}) + IN(t-1;(k-1)\{3\}) \\ &= IN(t-2;(k-1)\{2\}) + IN(t-2;(k-1)\{3\}) + IN(t-1;(k-1)\{3\}).\end{aligned}$$

3) $\hat{IN}(t;k)$: There are inflows into the current bucket k which we do not expect to have moved up to the next reporting bucket so they need to be added. Note that outflows only affect how much moves on to the *next* bucket but not how much sticks around:

$$\begin{aligned}\hat{IN}(t;k) \ &= \ \underbrace{IN(t;k\{1\}) + IN(t;k\{2\}) + IN(t;k\{3\})}_{IN(t;k)} \\ &\quad + IN(t-1;k\{1\}) + IN(t-1;k\{2\}) + IN(t-2;k\{1\}).\end{aligned}$$

Table A1—Effects of Assumptions on Flows

Total mass estimate	$O\hat{U}T(t;k-1)$	$I\tilde{N}(t;k)$	$I\hat{N}(t;k-1)$
Effect on excess mass	+	–	–
Max	Upper	Lower	Lower
Baseline	Upper	Upper	Upper
Min	Lower	Upper	Upper

4) $O\hat{U}T(t;k)$: Some of the mass that has moved up into the current bucket over the course of the past three months may have left the current bucket in the form of outflows, which we need to subtract. We only want to correct for the part that came in and then left again. So effectively we subtract the outflows from the bucket k from the inflow into bucket k. We cannot precisely tell which outflows exactly correspond to the inflows hence we just consider the total outflows. The earliest such outflow can occur at $t-1$. Outflows at time t do not affect the measure of excess mass: $O\hat{U}T(t;k) = OUT(t-1,k) + OUT(t-2;k)$.

We cannot measure the flows in and out of unobservable sub-buckets, which we denoted by $k\{1\}, k\{2\}, k\{3\}$. Hence we have to approximate the flows that we defined above by making an assumption how the total flow is distributed across the months that comprise a given bucket. We can however specify the bounds for each flow component and have an exact measure for the last. The bounds are as follows:

1) $0 \leq O\hat{U}T(t;k-1) \leq \sum_{j=0}^{3} OUT(t-j;k-1)$,

2) $0 \leq I\tilde{N}(t;k-1) \leq IN(t-1;k) + IN(t-2;k)$,

3) $IN(t;k) \leq I\hat{N}(t;k) \leq IN(t;k) + IN(t-1;k) + IN(t-2;k)$,

4) $O\hat{U}T(t;k) = OUT(t-1,k) + OUT(t-2;k)$.

Appendix Table A1 shows the combination of assumptions that generate the largest (and smallest) excess mass. For our baseline results, we choose the upper bounds for all flows which is a middle ground between combinations that yield that largest and smallest results respectively. In Figure A1 in the online Appendix, we present results using the maximum and minimum combinations respectively. Figure A1 in the online Appendix shows that the we get similar results when ignoring flows and simply using the formulas that only consider stocks. The reason is both that flows are small relative to stocks and that in many instances inflows and outflows cancel out.

The formulas above applied to three-month reporting buckets. The six-month buckets have analogous formulas.

A4. *Additional Restrictions*

We impose the following additional restrictions:

1) We impose that excess mass can never exceed observed mass in bucket.

2) We also impose that excess mass must be weakly positive since negative excess mass is just a mismeasured outflow: $B(t;k) \leq 0$.

3) We adjust for the common practice of banks to move overdue loans off their balance sheet in December to boost end-of-year statements, and putting the overdue balance back on in January. This leads to spurious fluctuation in our measure of excess mass.

Appendix B. Validity Checks for Algorithm

B1. *First Validity Test*

The first validity test regresses excess mass, the amount of underreporting, at firm i, bank b, month t, collateral type c, and reporting bucket k on a set of dummies that capture the increments in the mandatory deduction rate between reporting bucket k and $k + 1$:

$$\text{(B1)} \quad \frac{\textit{excess mass}_{ibkct}}{\textit{overdue loans}_{ibkct}} = \sum_{j=1}^{5} \beta_j \Delta \textit{deduction rate}_j + \varphi_b + \theta_i + \mu_t + \epsilon_{ibkct},$$

where i, b, c, k, and t index firms, banks, collateral type, reporting category, and month. We include firm-bank fixed effects and hence only use variation within a given lending relationship. We cluster standard errors at the firm-bank level; j indexes the possible increments in the regulatory rate, ranging from 0 to 25 percentage points.

The coefficient β_j measures the additional amount of excess mass that occurs in buckets when the change in the regulatory deduction rate from k and $k + 1$ is equal to $\Delta rate_j$, relative to buckets where the regulatory rate stays constant. If banks act strategically, we would expect all coefficients to be positive and statistically significant, and larger rate increments to have larger coefficients. We only consider relationships that have a single type of collateral to avoid confounding the estimate by including relationships with several types of collateral since the regulatory rules differ by collateral type. We estimate the specification separately for each type of collateral. Results are presented in Appendix Table B1 and discussed in Section I.

B2. *Second Validity Check*

Since we can directly trace the time a loan has been overdue in the subset of relationships with only a single loan, we can test whether underreporting is more pronounced when the regulatory deduction rate increases. For example, the regulatory rate increases when switching from reporting that the loan has been overdue

Table B1—Bunching at Points of Rate Increases

Excess mass/loans	No collateral (1)	Guarantee (2)	Real collateral (3)
Panel A. Bunching test			
Increase in deduction rate in next higher reporting bucket			
9 percentage points		0.244	0.110
		[0.002]	[0.005]
15 percentage points		0.451	0.349
		[0.002]	[0.004]
24 percentage points	0.178		
	[0.005]		
25 percentage points	0.324	0.098	0.014
	[0.005]	[0.001]	[0.002]
Observations	363,132	1,253,589	232,659
R^2	0.581	0.450	0.464
Panel B. Placebo test			
Increase in deduction rate in next higher reporting bucket			
25 percentage points	−0.024	−0.004	−0.015
	[0.001]	[0.001]	[0.001]
Observations	363,132	1,253,589	232,659
R^2	0.199	0.213	0.224

Notes: The table shows regression results for the first validity test of the loss underreporting algorithm. The dependent variable is the amount of excess mass (or underreporting) scaled by the total overdue loan balance in a given reporting bucket for a firm-bank relationship (see table 1 for a visual depiction). The explanatory variables are a series of dummies that capture how much the regulatory deduction rate increases from the current reporting bucket to the next higher reporting bucket (e.g., deduction rate of 1 percent versus 10 percent = increase of 9 percentage points). This difference measures the intensity of the incentive to underreport. The sample is split by collateral type since the regulatory rules differ by collateral type. Each column corresponds to the results of a regression in the sample of firm-bank pairs that only have that type of collateral. The omitted baseline category is 0 (no rate increase). Hence the coefficients capture how much more excess mass (or underreporting) occurs in reporting buckets where there is an increase in the regulatory rate in the next higher bucket. Regressions include firm × bank fixed effects. The placebo test regresses underreporting on buckets where there is a rate increase for the *other* collateral type but not for the given collateral type. Standard errors are clustered by firm-bank pair.

five months to reporting that is has been overdue six months. Hence the incentive to underreport is highest when the actual overdue duration has reached six months. By reporting that the six-month overdue loan continues to have been overdue only five months, the bank avoids the jump up in impairment losses associated with reporting six months. As in the first exercise we select the loans that have only a single type of collateral since the regulatory schedule differs by collateral type.

We confirm the existence of bunching by regressing the amount of excess mass in month t on a categorical variable that captures the same set of increments in the regulatory deduction rate as above. Appendix Table B2 confirms that an increase in the regulatory rate strongly correlates with an increase in the scaled amount of loss underreporting, or excess mass. For example, an increase in the rate by 24 percentage points leads to an 11 percentage point increase in the loan balance that is subject to loss underreporting (relative to the time periods without an increase in the regulatory rate). The effect is nonmonotonic with larger increases for the three to five

Table B2—Bunching in Sample of Single-Loan Relationships

Excess mass/loan balance	(1)	(2)	(3)
Increase in regulatory rate between $t-1$ and t			
9 percentage points	0.060	0.064	0.063
	[0.008]	[0.008]	[0.008]
15 percentage points	0.031	0.034	0.033
	[0.007]	[0.007]	[0.007]
24 percentage points	0.113	0.098	0.101
	[0.018]	[0.018]	[0.021]
25 percentage points	0.029	0.025	0.026
	[0.004]	[0.004]	[0.004]
Bank, firm fixed effects	Y	Y	N
Controls	N	Y	N
Observations	617,622	615,170	615,1702
R^2	0.016	0.115	0.115

Notes: The table shows regression results for the second validity test of the loss underreporting algorithm. The dependent variable is the amount of excess mass scaled by the total loan balance in a given reporting bucket for a firm-bank relationship (see Table 1 for a visual depiction). The explanatory variables are a series of dummies that capture how much the regulatory deduction rate increases from the reporting bucket in month $t-1$ to the reporting bucket at t; t refers to the constructed time overdue (counting in the data how long a loan has been overdue). The increase in the regulatory rate measures the intensity of the incentive to underreport. The sample only includes relationships with a single loan for which we can construct the time overdue. The omitted baseline category is 0 (no rate increase). Hence the coefficients capture how much more excess mass (or underreporting) occurs in months where there is an increase in the rate in the following month. Controls are the type of collateral. Standard errors are clustered by firm-bank pair.

months reporting category, which corresponds to $\Delta rate_{t-1} = 9$ for collateralized loans and $\Delta rate_{t-1} = 24$ for noncollateralized loans. This nonmonotonicity is due to the pressures to avoid classifying loans as nonperforming, explained in Section I.

Appendix C. Details on Productivity Decomposition

This Appendix reports equations used in the construction of the aggregate productivity decomposition in Section III.

C1. *Data and Variables*

We work with administrative data for nonfinancial firms as described in main text. We use the following variables:

Nominal output: value added (gross sales minus intermediate input costs)

Labor: number of employees

Capital: real capital stock. We deflate the stock of fixed assets in 2006 (or the earliest available firm-level observation) by the 2006 capital goods deflator. We then compute the firm-level change in real fixed assets by adjusting lagged real fixed assets by the firm-level depreciation rate and adding firm-level investment spending according to the following formula: $k_{it} = \left(1 - \Delta_{it}\right)k_{t-1} + \left(I_{it}/def_t\right)$ From 2009 onward, we use capital expenditure (CAPAX) reported in the cash-flow statement when available (which is expenditure on tangible and intangible investment). Before 2009, or when CAPEX is not reported, we simply use the change in the book value of fixed assets. We deflate investment spending by the capital goods deflator.

Sectoral labor share: one minus the ratio of sectoral gross operating surplus over value added.

All sectoral variables are at the two-digit level.

C2. *Setup*

Each firm i in sector s produces a differentiated good using a Cobb-Douglas production function:

$$Y_{it} = A_{it} K_{it}^{\alpha_s} L_{it}^{\beta_s},$$

where Y_{it} is value added, L_{it} and K_{it} denote labor and capital, and A_{it} is firm-level TFP. We define the constant returns to scale parameter $\gamma_s \equiv \alpha_s + \beta_s$.

Output in each sector s is given by a CES aggregate:

$$Y_{st} = n_{st}^{\frac{\theta_s - 1}{\theta_s}} \left(\sum_{i \in \mathcal{N}_{st}} Y_{it}^{\theta_s} \right)^{1/\theta_s},$$

where the elasticity of substitution within sector s being defined as $1/(1 - \theta_s)$ with $0 < \theta_s < 1$.

Aggregate output is then given by $Y_t = \left(\sum_s Y_{st}^{\rho}\right)^{1/\rho}$ where the elasticity of substitution across sectors being defined as $1/(1 - \rho)$ with $0 < \rho < 1$.

C3. *Decomposition*

We use the following exact decomposition following Osotimehin (2019). We ignore the entry/exit component and work only with the sample of continuing firms.

Technical efficiency (TE) is computed as

$$\Delta TE \simeq \sum_{s=1} \frac{1}{1 - \gamma_s \rho} \left[\nu_{st,t-1}^{Y} - \rho \epsilon^K \nu_{st,t-1}^{K} - \rho \epsilon^L \nu_{st,t-1}^{L} \right] \ln I_{TE},$$

where

$$\nu_{st,t-1}^{Y} \equiv \left(\frac{P_{s,t-1} Y_{s,t-1}}{P_{t-1} Y_{t-1}} + \frac{P_{st} Y_{st}}{P_t Y_t} \right) / 2,$$

and ν^K and ν^L are similarly defined.

The elasticity parameters are defined as the time t and $t-1$ averages of

$$\epsilon^K = \frac{\alpha^Y - \rho \alpha^L}{1 - \rho(\alpha^L + \beta^K)}$$

and

$$\epsilon^L = \frac{\beta^Y - \rho\beta^K}{1 - \rho\left(\alpha^L + \beta^K\right)},$$

where $\alpha^Y = \sum_s \alpha_s(P_s Y_s)/(PY)$, with β^Y similarly defined, and $\alpha^L = \sum_s \alpha_s(L_s/L)$, with β^K similarly defined.

Sectoral technical efficiency changes are defined as

$$\ln I_{TE} \equiv \ln\left(\sqrt{I_{TE0}}\sqrt{I_{TE1}}\right),$$

with

$$I_{TE0} = \left(I_{TE0}^Y\right)^{1/\theta}\left(I_{TE0}^K\right)^{-\alpha_s}\left(I_{TE0}^L\right)^{-\beta_s},$$
$$I_{TE1} = \left(1/I_{TE1}^Y\right)^{-1/\theta}\left(1/I_{TE1}^K\right)^{\alpha_s}\left(1/I_{TE1}^L\right)^{\beta_s}.$$

These indexes in turn are defined as

$$I_{TE0}^Y = \sum_i \left(\frac{A_{it}}{A_{i,t-1}}\right)^{\frac{\theta_s}{1-\gamma_s\theta_s}} \frac{p_{i,t-1}Y_{i,t-1}}{\sum_i p_{i,t-1}Y_{i,t-1}},$$
$$1/I_{TE1}^Y = \sum_i \left(\frac{A_{i,t-1}}{A_{it}}\right)^{\frac{\theta_s}{1-\gamma_s\theta_s}} \frac{p_{it}Y_{it}}{\sum_i p_{it}Y_{it}}.$$

Changes in within-sector allocative efficiency (AE) are given by

$$\Delta AE_{within} \simeq \sum_s \frac{1}{1-\gamma_s\rho}\left[\nu_{st,t-1}^Y - \rho\epsilon^K\nu_{st,t-1}^K - \rho\epsilon^L\nu_{stst-1}^L\right]\ln I_{AE},$$

where $I_{AE} \equiv \sqrt{I_{AE0}}\sqrt{I_{AE1}}$ varies at the sector-level and the indexes are similarly defined as in the technical efficiency case above.

Finally, changes in between-sector allocative efficiency are defined as

$$\begin{aligned}\Delta AE_{between} \simeq &\sum_{s=1}\frac{1}{1-\gamma_s\rho}\left[\epsilon^K\left(1-\beta_s\rho\right)\nu_{st,t-1}^K + \epsilon^L\alpha_s\rho\nu_{st,t-1}^L - \alpha_s\nu_{st,t-1}^Y\right]\\ &\times\frac{\Delta\omega_{st}^K}{1+\omega_{st-1}^K} + \sum_{s=1}\frac{1}{1-\gamma_s\rho}\left[\epsilon^K\beta_s\rho\nu_{st,t-1}^K + \epsilon^L\left(1-\alpha_s\rho\right)\nu_{st,t-1}^L\right.\\ &\left. - \beta_s\nu_{st,t-1}^Y\right]\frac{\Delta\omega_{st}^L}{1+\omega_{st-1}^L},\end{aligned}$$

where we measure sectoral wedges using sectoral marginal products: $\omega_{st}^x = MVP_{st}^x$ for $x = K, L$.

Aggregate TFP changes in the data (the left-hand side of the decomposition) are computed as

$$\Delta\log TFP = \Delta\ln Y_t - \epsilon^k\Delta\ln K_t - \epsilon^L\Delta\ln L_t.$$

Firm-level TFP is computed as

$$A_{it} = \frac{\left(p_{it} Y_{it}^{1/\theta_s} / P_{st}\right)}{K_{it}^{\alpha_s} L_{it}^{\beta_s}} \left(\frac{Y_{st}}{n_{st}}\right)^{(\theta_s - 1)/\theta_s},$$

where we used the following assumption about the demand function: $p_{it}/P_{st} = \left(n_{st} Y_{it}/Y_{st}\right)^{\theta_s - 1}$.

We set the returns to scale parameter to $\gamma_s = \gamma = 1$. We set $\theta_s = \theta = 0.66$ and $\rho = 0.5$. This corresponds to an elasticity of substitution of 3 within sectors and of 2 between sectors.

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